SUSTAINABILITY
TRENDS

TANKERS

ABS

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INTRODUCTION

OVERVIEW

Emerging energy regulations and ongoing industry studies into options for reducing the emissions from ships are progressively stimulating innovation and preparing companies for a new wave of technologies.

The International Maritime Organization's (IMO) initial greenhouse gas (GHG) strategy was adopted in April 2018, establishing three levels of ambition that are subject to ongoing reviews by the organization. These set mandates to ensure:

- The carbon intensity of newly constructed ships declines by further developing regulations known as the Energy Efficiency Design Index (EEDI);
- A decline in carbon intensity of international shipping; and
- The GHG emissions from international shipping will peak, and then decline.

The ambition levels are designed to meet specific targets that support the global efforts against climate change. Projecting into the near and distant future, the strategy's goals aim to curb carbon dioxide (CO_2) emissions by 40 percent by 2030 (against 2008 levels) and pursue a 70 percent reduction by 2050; GHG emissions are also expected to peak, and then fall by 50 percent by 2050.

Because the IMO's intersessional working group on GHG reduction (ISWG-GHG) has identified improvements in vessel design, operational performance and the immediate introduction low- and zero-carbon fuels as ways forward, the ambition levels are seen as goal-based principles on which the global shipping sector can act.

A list of short-, medium- and long-term measures to support the IMO's ambition levels was also agreed. Short-term measures include the evaluation and improvement of the energy-efficiency requirements for ships (EEDI, Ship Energy Efficiency Management Plan [SEEMP] regulations), the application of technical-efficiency measures for existing ships (from regulations known as the Energy Efficiency Existing Ship Index [EEXI]) and the introduction and regulation of carbon intensity (known as the Carbon Intensity Indicator [CII]) for ships in operation.

The medium- and long-term measures include developing a program to implement alternative low- and zero-carbon fuels, the adoption of other innovative emissions-reduction mechanisms, and market-based measures (MBMs) to incentivize the reduction of GHG emissions.

Depending on the operational profile and age of their tankers and fleet-replacement strategies, owners are expected to reach compliance by combining current options – such as limiting engine power and vessel retrofits – with emerging options such as alternative fuels, battery propulsion or fuel cells.


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EEDI COMPLIANCE IN THE TANKER SECTOR

ABS has examined the datasets reported to the IMO for EEDI-verified tankers. The cumulative summary published in March 2021 on the IMO Global Integrated Shipping Information System (GISIS) provides a graphic illustration of the efficiency levels for tankers at that time (see Figure 1 below).

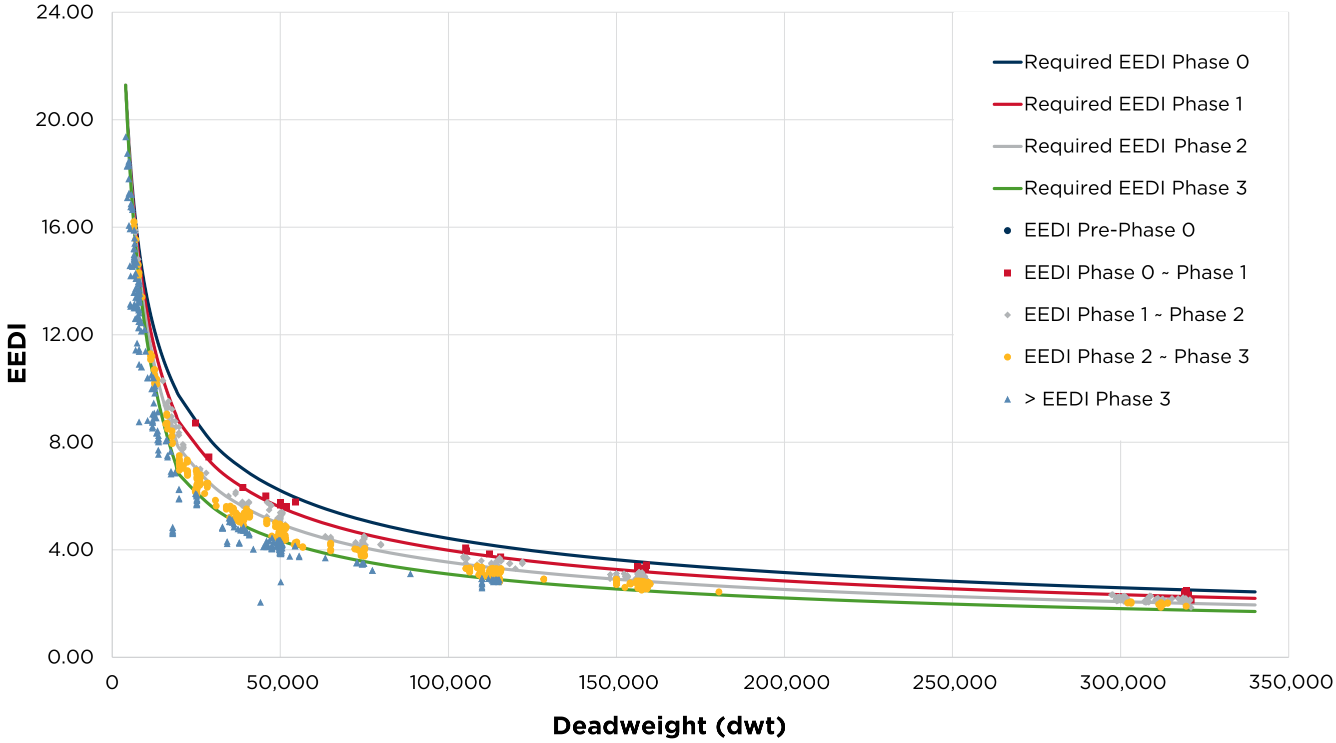


Figure 1: Tankers' EEDI compliance overview.

The EEDI regulations were introduced to optimize ship designs within a formal framework, while simultaneously stimulating innovation. However, the improvement trends are not uniform across the standard commercial sizes (e.g., medium range [MR], aframax, very large crude carrier [VLCC]).

In part, this is because the population of ships used for the regression was not equally distributed across the industry sector when the reference line for tankers was developed by the IMO Marine Environment Protection Committee (MEPC) working group. Some deadweight (dwt) segments (50k dwt, for example) weighed more on the creation of the baseline curve compared to others; the discrepancy is greater for vessels of larger capacity.

To provide a fair assessment of the evolution of tanker designs since the EEDI was implemented, and to strengthen the regulations, the different approaches to design taken by shipyards and designers would need to be documented.

However, the IMO database does not provide references to compare any efforts to optimize hull forms, modify structures or implement devices that improve propulsion. The latter, in particular, are very common to designs delivered after 2013 (e.g. pre-swirl stators, rudder bulbs) and are often coupled with optimized propeller and rudder configurations. Their efficiency benefits cannot be separated from the overall performance of the ship, as they are accounted for in the EEDI reference speed (VREF) during model tests and speed trials.

Meanwhile, the IMO MEPC has repeatedly raised concerns about the lack of uptake for innovative technologies (fourth term which contribute to the ship's propulsion needs and fifth term that generate electricity) by tankers, in particular, and other ship types. It should be noted that the innovative technologies applied to the few VLCC tankers that have been verified for EEDI and reported to the IMO GISIS are fourth term.

The most common fourth-term technologies fitted on commercial cargo vessels are the waste-heat recovery systems used to generate electric power. A direct comparison of VLCC tankers with identical main parameters, capacity, EEDI power (power of main engines [PME], generally defined as 75 percent of the ship engine's installed maximum continuous rating) and corresponding VREF, indicates that energy-efficiency benefits range between 1.9 and 2.7 percent.

As the oil-tanker industry enters the final stretch towards EEDI Phase III (Jan. 1, 2025), significant improvements will be necessary for new designs to reach compliance. This may prove a challenge, particularly for larger vessels where the limits already have been met for engine derating, as governed by the IMO's minimum propulsion power regulations.

EEXI REGULATIONS

To support the first level of ambition for the efficient ship designs envisioned in the IMO's GHG strategy, the organization adopted the proposed EEXI regulations in June 2021; this introduced requirements for the operational fleet and aligned them with the EEDI regulations for ships under construction.

The EEXI will need to be calculated for each ship and benchmarked for compliance against the EEDI baseline (Phase 0). The reduction factor (percent below that baseline) was initially set at 20 percent for all commercial tanker sizes.

In recognizing the challenges that the larger dwt segments of new designs will have under the EEDI scope and considering that older hulls would face an even greater challenge, the IMO recently agreed to adjust the target to 15 percent for all ships of 200k dwt and above.

This can be seen in Figure 2 (below), where ABS has examined the IMO reported datasets for EEDI verified tankers to evaluate compliance with the EEXI limit.

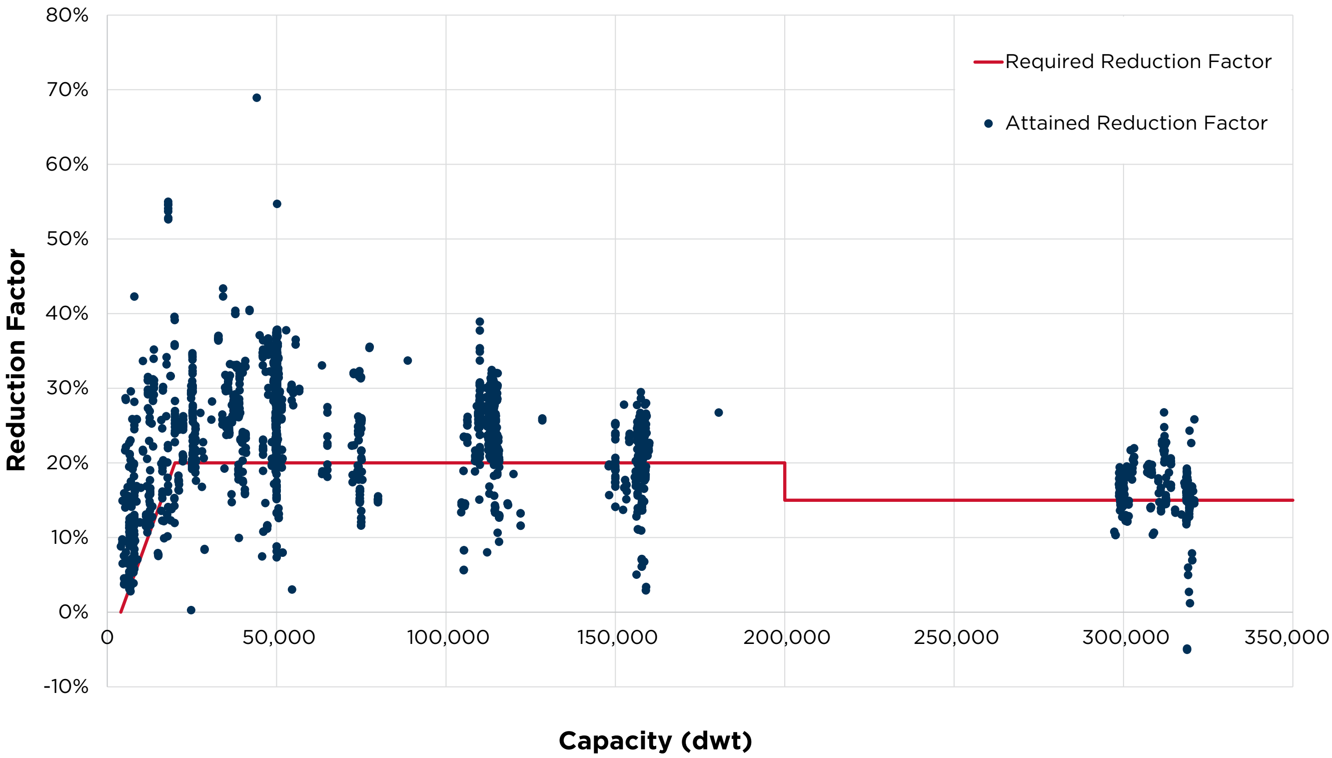


Figure 2: Tankers' EEXI compliance overview.

As illustrated, even the newer ship designs will require modifications to meet the EEXI limits. This most likely will result in the adoption of solutions such as limits on engine or shaft power, slowing the ships' transit speeds. Of course, the crew will be able to override those limitations in cases of emergency (e.g. piracy, search and rescue operations), or when the ship encounters adverse weather.

Commercial contracts may need to be reevaluated based on the regulatory developments to reach a balance between compliance and the demands of trade.

THE CII COMPLIANCE TRAJECTORY

The annual requirements for governing the carbon intensity of ships were also adopted by the IMO in June 2021, addressing the GHG strategy's second level of ambition. The CII mechanism represents the IMO's first effort to evaluate and benchmark the performance of ships in service.

Using each vessel's annual fuel consumption reported under the IMO data collection system (DCS), each vessel's attained CII will be calculated and benchmarked against the regulation's reference line using an annual-reduction rate (Z). Similar to the scope of the EEXI, the reference line is a dwt-based regression using data from the fourth IMO GHG study.

The reduction rate Z for tankers (and for other ship types) has yet to be agreed, but it will most likely reside between 0.5 percent (a demand-based approach) and two percent (supply-based approach).

Vessels that attain an annual CII within a certain bandwidth (above or below the reference line) will be issued a rating from A to E. Compliance will be determined by whether the ship maintains an annual rating of C or better.

Figure 3 shows the components that comprise the CII mechanism; some of the parameters are still under discussion at the IMO.

ATTAINED CII

$$CII = \frac{CO_2\ Emissions}{Transport\ Work} = \frac{Fuel\ Consumption\ x\ Cf}{DWT\ x\ Distance}$$

REQUIRED CII

$$CII_{ref} = a * Capacity^{-c}$$

ANNUAL CII REDUCTION

$$CIIreq,\ year = CII_{ref}\left(1 - \frac{z}{100}\right)$$

CII RATING BOUNDARIES

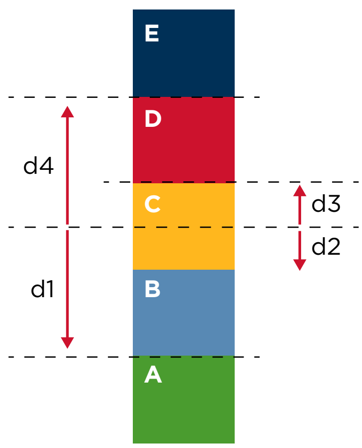


Figure 3: The CII mechanism.

Compliance with EEXI regulations, which is a one-off compliance certificate, will be the starting point for tankers on their journey to meeting the annual targets for CII, which are strengthened annually. This means that, even if a vessel is currently awarded a higher rating (e.g., A or B), adhering to a "business-as-usual" mindset will not be enough for fleet operations to remain sustainable (see Figure 4).

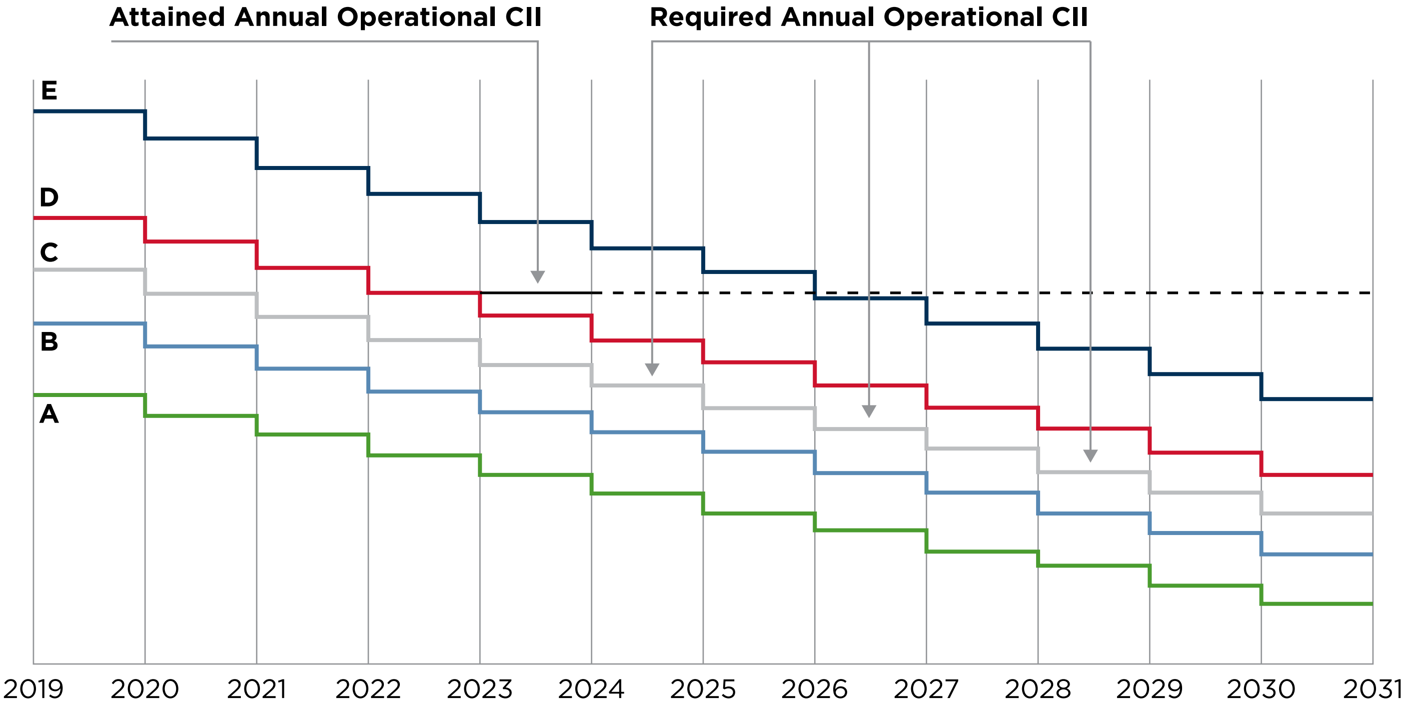


Figure 4: The CII trajectory toward 2030 compliance.

Therefore, it is critical that no stone is left unturned in the pursuit of improved efficiency. Shipyards and designers are exploring designs that use alternative fuels to power all or part of the vessel's cruising range. Fuel-storage capacity, which could reduce a vessel's dwt, presents a hurdle due to the combination of energy content and low density found in many low- and zero-carbon fuel solutions.

Hybrid propulsion systems (which combine an internal-combustion engine with batteries) are also options to explore, although these configurations may be limited to the short-haul trades.

Even though slowing vessel speeds is a theoretical solution, older tonnage will prove challenging from a market perspective, given that these full hull forms are already sailing at speeds much lower than their design point, mostly due to commercial requirements for fuel savings.

MARKET OVERVIEW

The COVID-19 pandemic has severely impacted the seaborne trade in oil, with the crude trade falling by eight percent year-on-year in 2020; the product trade fell 12 percent.

While there were small signs of recovery as the calendar turned to 2021, the seaborne trade in oil and products still remained down nine and six percent in the first quarter, respectively.

By most accounts, the downward pressure on the oil trades was driven by lower energy demand during the pandemic as the global economy partially shut down, high pre-pandemic oil inventories, reduced output from Organization of Petroleum Exporting Countries (OPEC+) and weak refinery throughput.

After the global reserves held in floating-storage assets (these reserves reached their height from March to April 2020) were depleted, market pressure built in the second half of 2020 and continued to be weak in the first quarter of 2021.

However, tanker demand is projected to improve gradually in the short term, with the demand in seaborne oil trade expected to return to 2019 levels by 2022.

Longer term, this expectation will need to be balanced with the world's overall transition to lower- and zero-carbon fuels, as capacity demands in some tanker segments are among shipping's most vulnerable to that transition and the associated regulations.

As the world acts to reduce carbon consumption, this transition will have implications on demand for capacity across specific segments of the global tanker fleet for the foreseeable future.

For example, in 2019, crude oil accounted for approximately 17 percent of the total goods shipped by sea. However, the latest BP Energy Outlook forecasts world consumption (under its "rapid" scenario) to fall by 500,000 barrels per day each year until 2030.

This will clearly have an impact on tanker demand and, for some industry-watchers, BP's outlook is a conservative view of the pending decline, which is expected to accelerate past 2030.

From 2005 to 2015, the overall global tanker fleet (comprised of crude, product and chemical tankers) expanded by an average of 4.4 percent every year, as measured in gross tons (gt). Since 2015, that growth has slowed to 3.8 percent (see Figure 5 below).

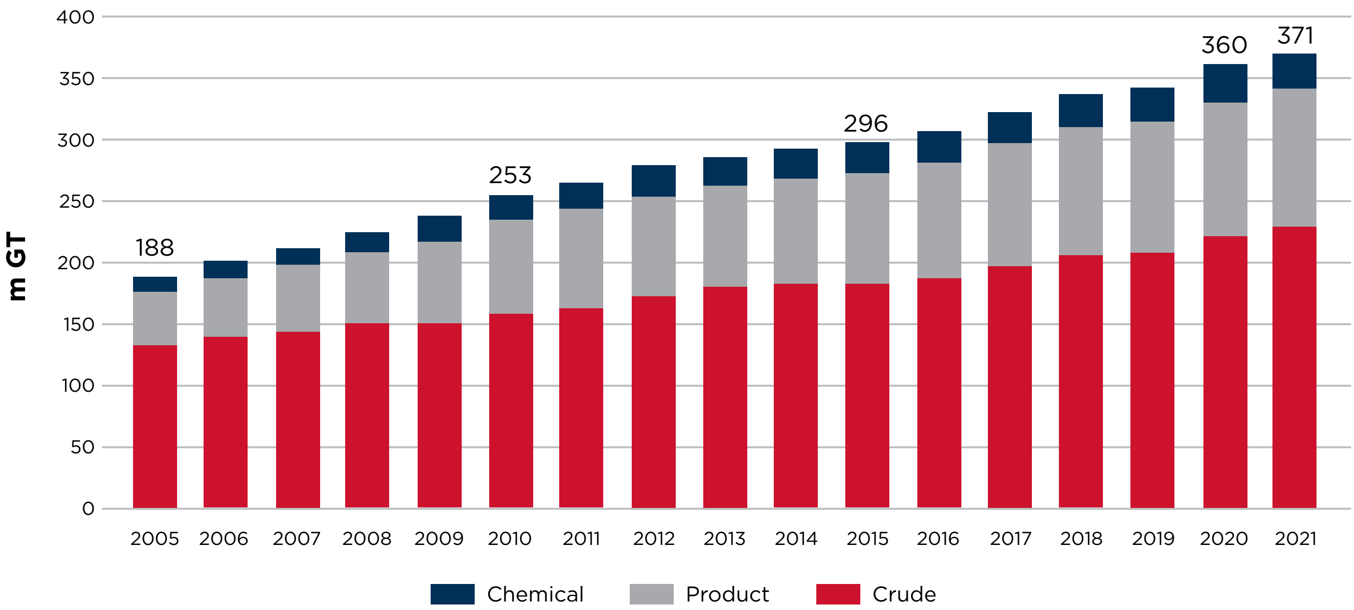


Figure 5: Tanker fleet development.

Since 2005, the capacity of the crude tanker fleet experienced the slowest compound annual growth rate (CAGR), at 3.6 percent (see Figure 6), of three main tanker segments.

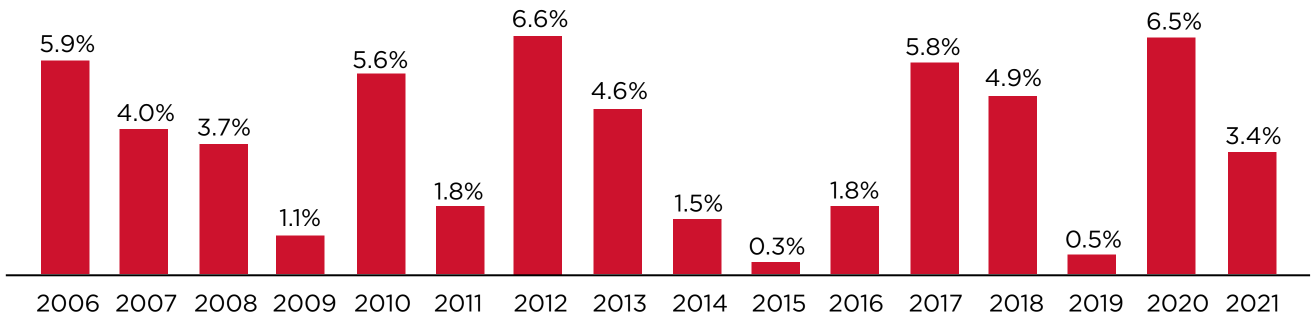


Figure 6: Growth in crude tanker fleet.

At a 5.7 percent CAGR, the fleet of chemical tankers experienced comparatively better growth in gt over the same period (from 2005), with healthy demand for petrochemical and longer trading distances (ton-mile growth) driving an uneven demand for new tankers (see Figure 7 below).

While the trade in seaborne chemicals dropped two percent in 2020 due to the impact of COVID-19, the next two years are expected to rebuild demand for their movement, driven by renewed demand for petrochemicals and the global repositioning of refineries.

For the next two-year period, expectations are for the chemical segment to see a four percent CAGR in seaborne trade with a one percent growth in supply potentially offering a short-term market balance.

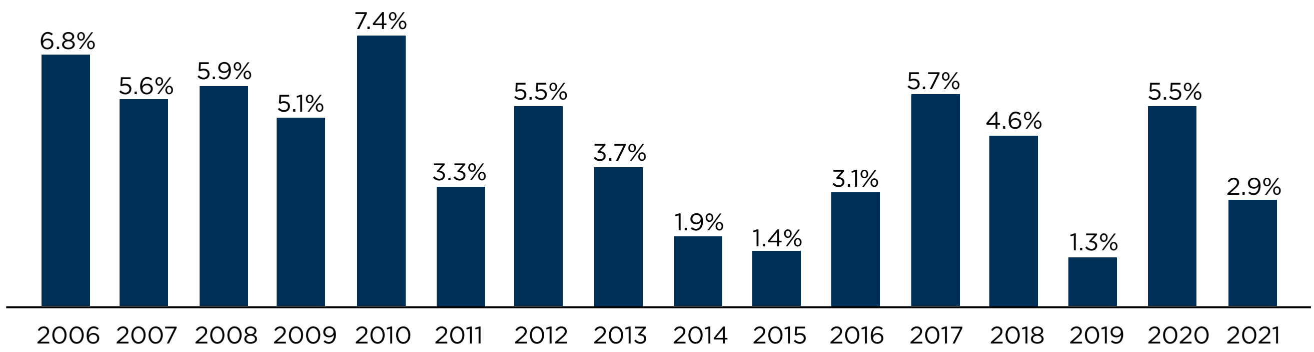


Figure 7: Year-on-year growth of chemical tanker fleet.

Tanker shipping's strongest growth is being witnessed by the owners of the product fleet, which has seen a CAGR of 5.8 percent since 2005, although new capacity was injected into the fleet at a slower rate after 2010 (see Figure 8 below).

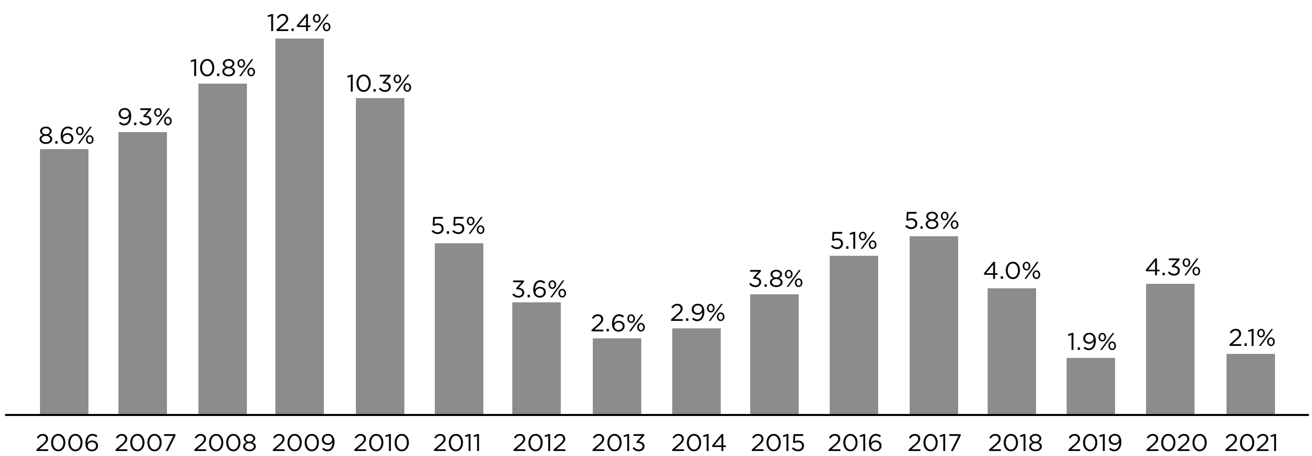


Figure 8: Year-on-year growth of the product tanker fleet.

There is some ongoing repositioning of refinery capacity closer to the current demand markets. But the overall development of new export hubs for the product trade is expected to alter trade patterns and lengthen ton-miles on the segment, ostensibly creating demand for future fleet expansion.

Another potential driver for fleet renewal is its age; some 28.5 percent of product carriers are 15 years of age or older. This is the result of two factors: 1) owner reluctance to invest in newbuilds; and 2) the emergence of more charterers who will accept vessels in that age range.

However, this trend is likely to reverse in the short-to-medium term due to incoming environmental regulation (EEXI, CII, for example) and the fact that a short-term surplus of capacity is expected to drive an increase in scrapping throughout 2021 and 2022.

AGE OF THE TANKER FLEET

In general, all tanker segments are presently exhibiting relatively low ratios comparing existing orderbook to ships over 20 years old, suggesting that the overall supply and demand for capacity is in relative balance.

On average, the global crude fleet, which numbers just over 2,200 vessels, is the youngest among tankers segments (see Figure 9 below) and features the lowest proportion of ships over 20 years of age. Its orderbook-to-ships-over-20-years-old ratio is 8.4 to 8.5 percent, respectively, suggesting balance.

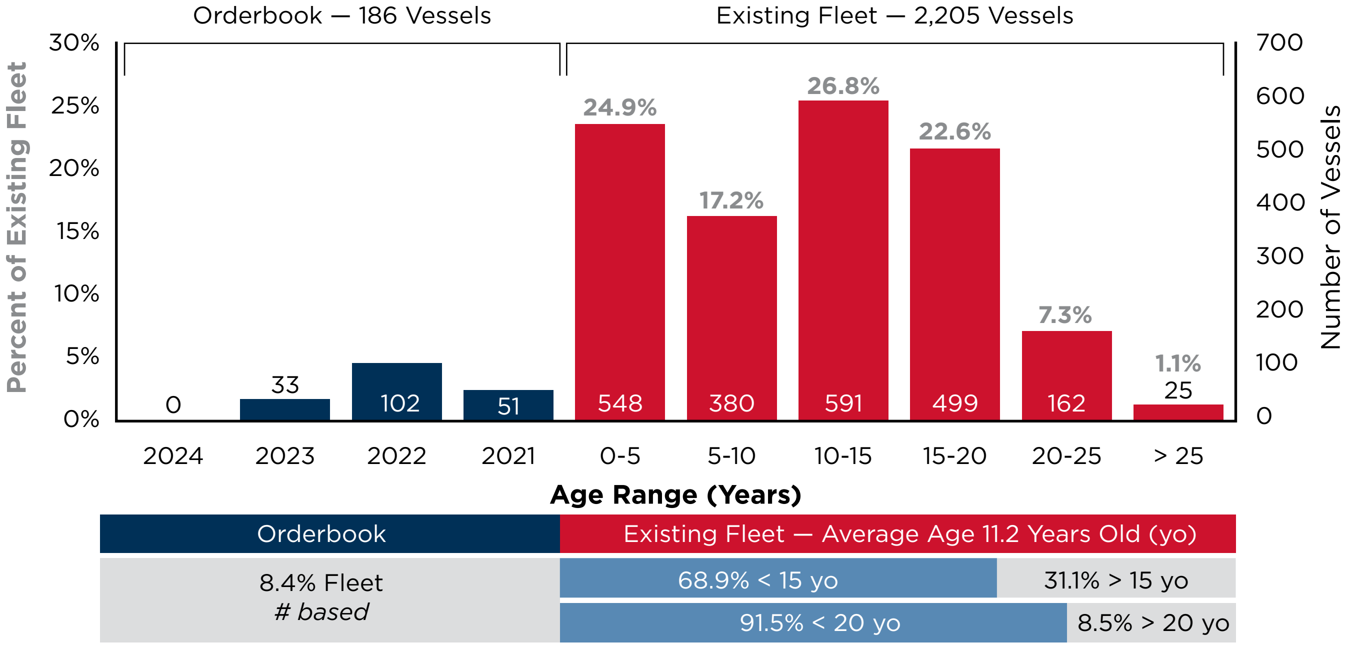


Figure 9: Crude fleet demographics.

However, almost 23 percent of the existing crude fleet is in the 15 to 20 year old range at a time when the charterers' and oil majors' environmental, social and corporate governance demands are ramping up. Despite the balanced ratio, further fleet renewal is expected in the short term as the scrapping age for tankers trends down.

The chemical fleet offers the oldest average age among the tanker segments, with just over a fifth of ships in the segment having surpassed 20 years of age (see Figure 10). With an orderbook-to-ships-over-20-years-old ratio of 4.5 to 20.7 percent, the segment would appear to have an opportunity for fleet renewal without significantly affecting the supply-demand balance. However, for the reasons discussed above pertaining to owner's reluctance to order and charterer comfort with older ships, the orderbook presently boasts just 163 vessels.

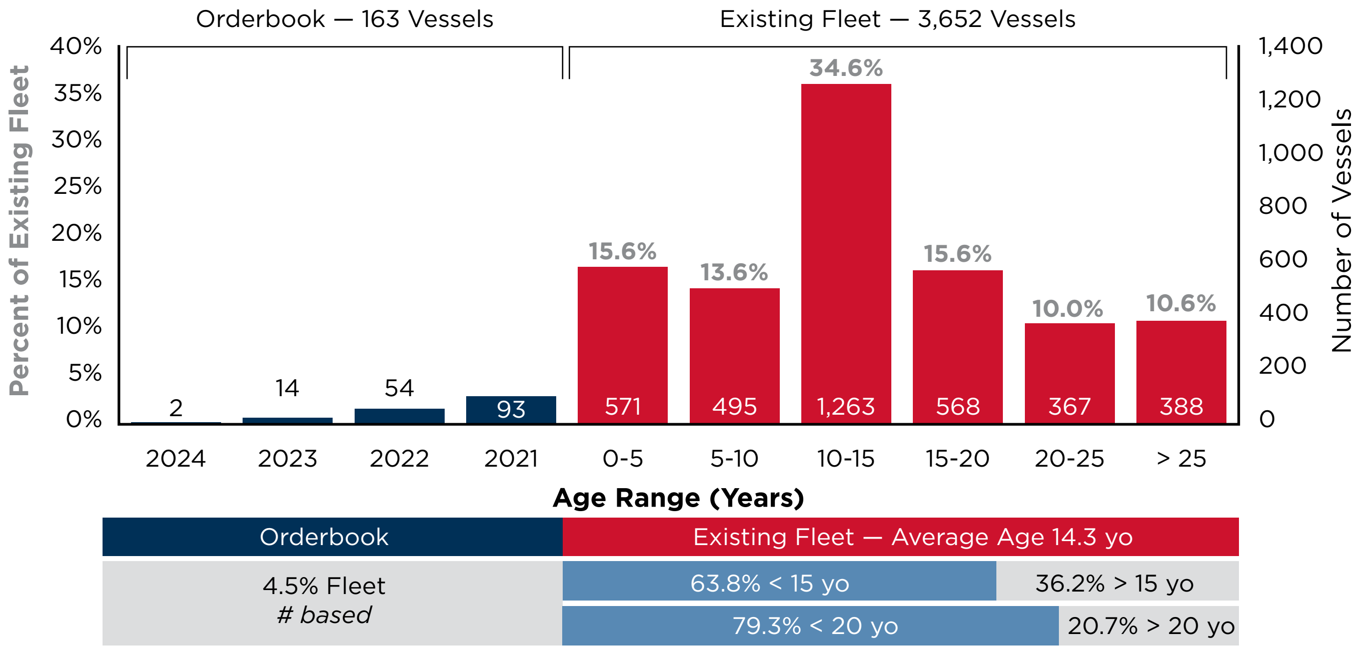


Figure 10: Chemical fleet demographics.

Whether new corporate social responsibility initiatives such as the Sea Cargo Charter, which forms a global framework for assessing and disclosing the environmental performance of chartering activities, will provide greater impetus for renewal in this segment remains to be seen.

At an average of 11.8 years, the global product-tanker fleet is suited to renewal despite it having the highest compound capacity growth of the three segments since 2005 (see Figure 11 below).

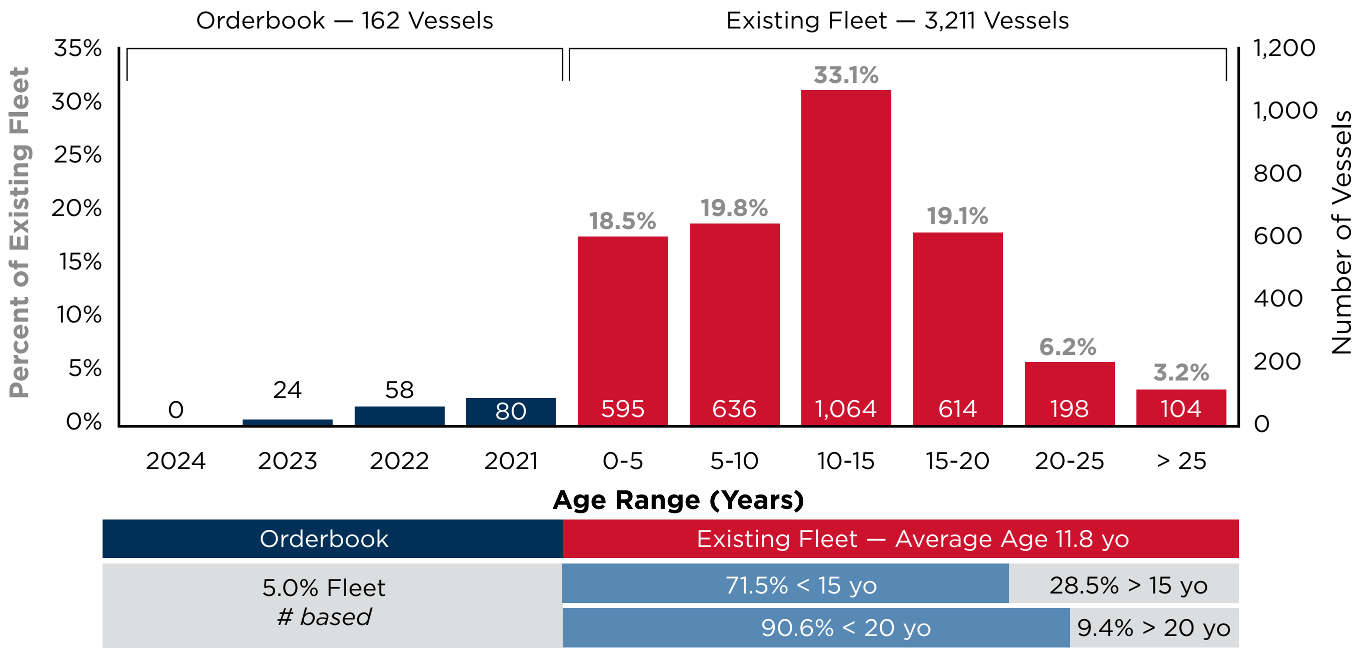


Figure 11: Product fleet demographics.

Just over 28 percent of the 3,211-unit strong product fleet has surpassed the age of 15, and its orderbook-to-20-years-old-ship ratio is five to 9.4 percent, suggesting the conditions exist for an upgrade without significantly disturbing capacity supply-demand balance.

Renewal is expected to accelerate in the short term, especially in the smaller segments where the average unit age is 14.3 years. This fleet also disproportionately operates in the short-sea sector, where renewal and scrapping is increasingly influenced by having to perform in Environmental Control Areas, where regulation is more stringent.

SCRAPPING AND DELIVERIES

Since 2005, an annual average of 2.7 million (m) gt of crude tankers has been scrapped; the heights for that period were reached in 2017 (4.8m gt) and 2018 (9.1m gt [see Figure 12 below]).

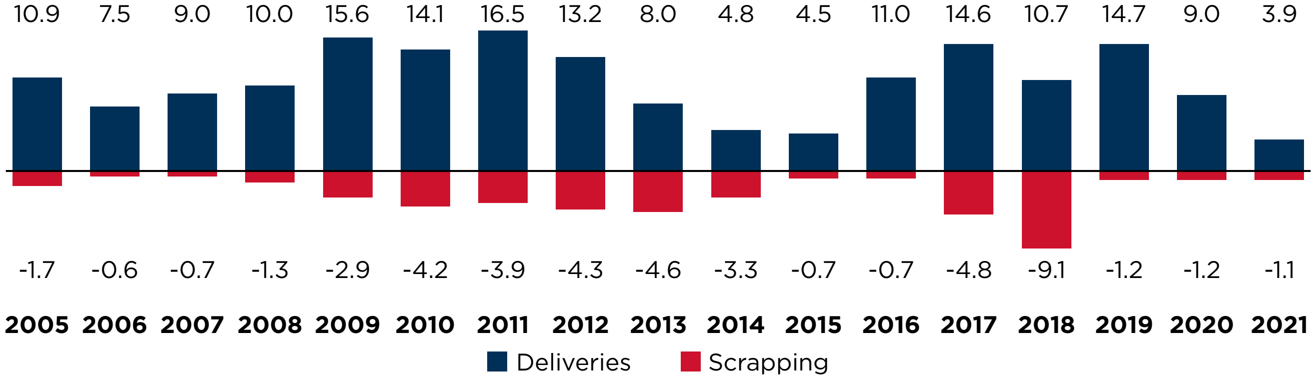


Figure 12: Crude tanker deliveries and scrapping since 2005.

During this period, annual crude tanker deliveries averaged about 10.4m gt, creating the youngest fleet of the segments being examined.

The chemical tanker fleet, the oldest among the segments being examined, scrapped an average of just 300,000 gt a year during the period, for reasons discussed in the sections above (see Figure 13 below).

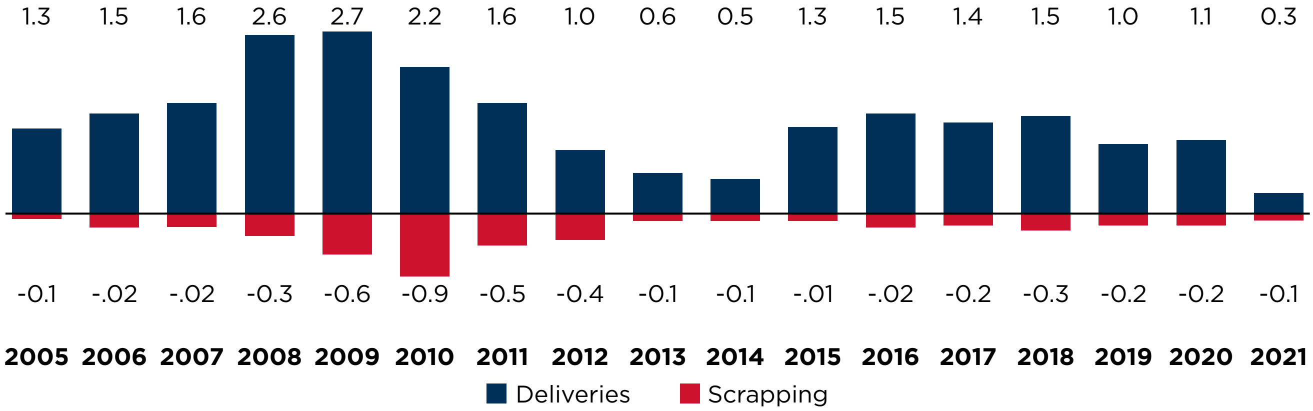


Figure 13: Chemical tanker scrapping and deliveries since 2005.

While the global chemical fleet added an average of about 1.4m gt a year in new ships, underpinning a relatively strong expansion, the fleet's ratio of orderbook to older ships leaves room for further renewal.

Finally, the strong capacity growth seen by the product tanker fleet since 2005 was underpinned by an additional 5.4m gt in new tonnage every year versus an average annual scrapping rate of just 1.1m gt (see Figure 14).

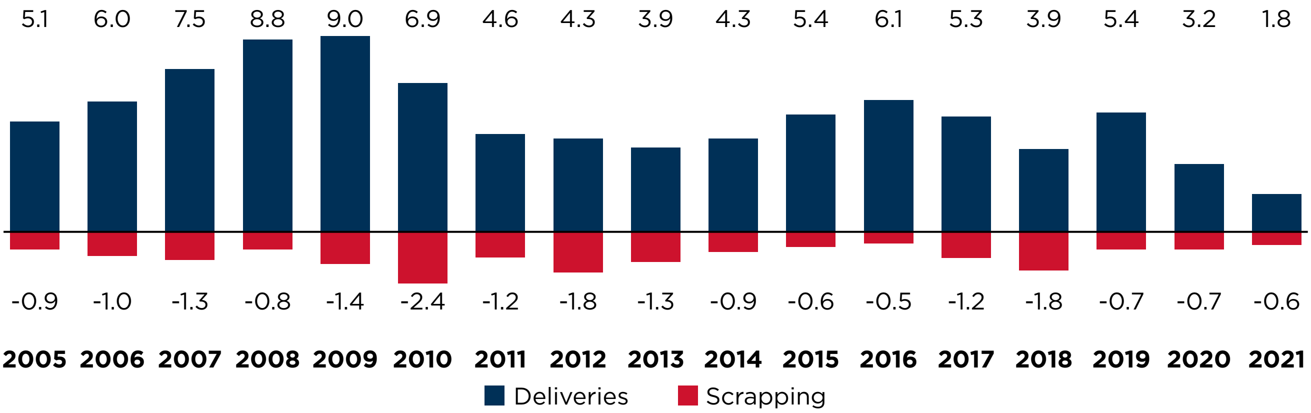
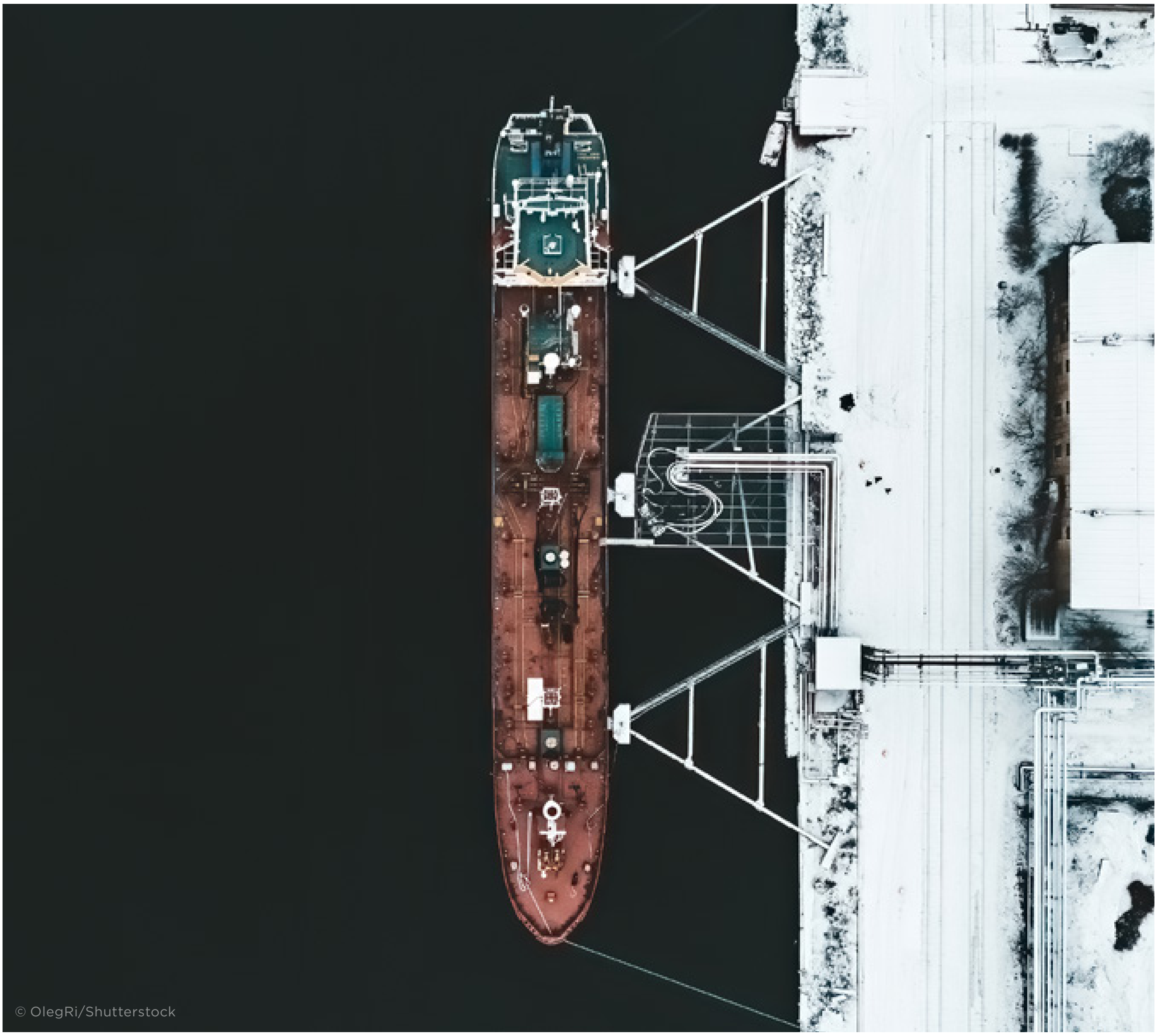


Figure 14: Product tanker scrapping and deliveries since 2005.

Overall, the global tanker fleet has the underlying conditions that would appear to support a measured renewal without disrupting the market.

The pace at which new efficiency-promoting technologies are adopted is expected to accelerate, supporting renewal efforts just as corporate social responsibility initiatives from tanker financiers and charterers gather steam and new regulations such as the EEDI, EEXI and CII enter into force.



CURRENT EVOLUTION OF SHIP DESIGNS

This section reviews the current state of the tanker fleet and the associated activities intended to improve its environmental sustainability across three main sectors: MR, aframax and suezmax. In general, it reveals measured progress in improving the design-based efficiency of these types of ships, a pace that can be attributed to the relatively slow adoption of innovative technology.

The requirements for the IMO's 2020 global sulfur cap came into force on January 1, 2020. For tanker vessels, the most commonly adopted compliance options have been the use of sulfur-compliant fuels or the installation of sulfur oxide (SO_x) scrubbers. The adoption of these solutions is discussed across three vessel-capacity sectors: MR (dwt ≥ 25k but < 60k); aframax (dwt ≥ 80k but < 120k); and suezmax (dwt ≥ 120k but < 200k).

According to Clarksons Research, as of mid-February 2021, 415 MR, aframax and suezmax tankers – both operational and on-order – had been fitted/were being fitted with SO_x scrubbers (see Figure 15 below); suezmax tankers have had the greatest proportional adoption of SO_x scrubbers.

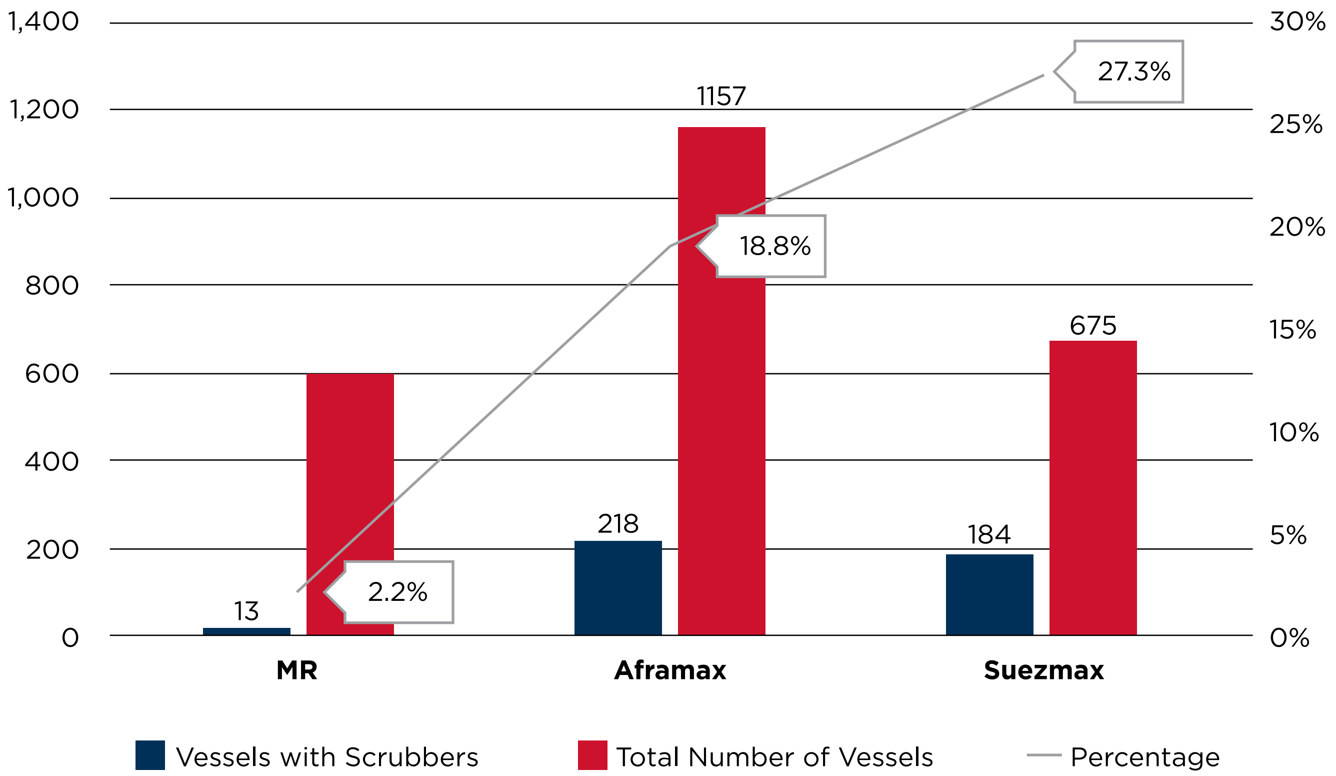


Figure 15: At 27.3 percent, the suezmax fleet has seen the greatest proportional uptake of SO_x scrubbers.

While the number of vessels adopting SO_x scrubbers had grown to more than 4,700 as of February, it remained just a small portion of the world's fleet, accounting for about eight percent of vessels greater than 2,000 dwt. Most of the global fleet had opted for sulfur-compliant fuel to meet the IMO 2020 requirements.

The adoption of sulfur-compliant fuel has not been without challenges; however. Since 2020, reports of engine problems have highlighted issues such as worn cylinder liners, broken piston rings and injector failures; early indications are that the quality/compatibility of some of the new fuels are the culprits.

There are many types of very low sulfur fuel oils (VLSFO) produced from blending processes and feed stocks. This wide variety poses quality challenges associated with their properties, including:

- Compatibility
- Stability
- Catalytic (Cat) fines
- Density
- Flash point
- Ignition and combustion characteristics
- Unusual components
- Pour points
- Viscosity
- Compatible lubricants

Simply put, unless the fuels are properly treated to the technical requirements of the combustion units in which they are intended for use, fuel properties have the potential to damage marine engines.

One such property is Cat fines, which mostly appear in fuels where catalysts are used for catalytic cracking during the refining processes from crude oils. Engine manufacturers usually recommend limits and sizes for Cat fines for the fuels supplied to specific engines. If the limit or size exceeds the recommended value, damage to cylinder lines or piston rings is possible.

An incompatibility of lubricants and fuels also can cause cylinder wear and damage to piston rings. When a fuel's sulfur content is reduced, it typically provides less lubrication than the high-sulfur varietals. Therefore, following the instructions of the engine manufacturer to select the lubricants with the appropriate base number is important. Overall, it is important to manage, treat and handle fuels in accordance with their properties.

FLEET PROFILES

As of mid-February 2020, the global fleet (in service and on order) had reached 600 MR, 1,157 aframax and 675 suezmax tankers.

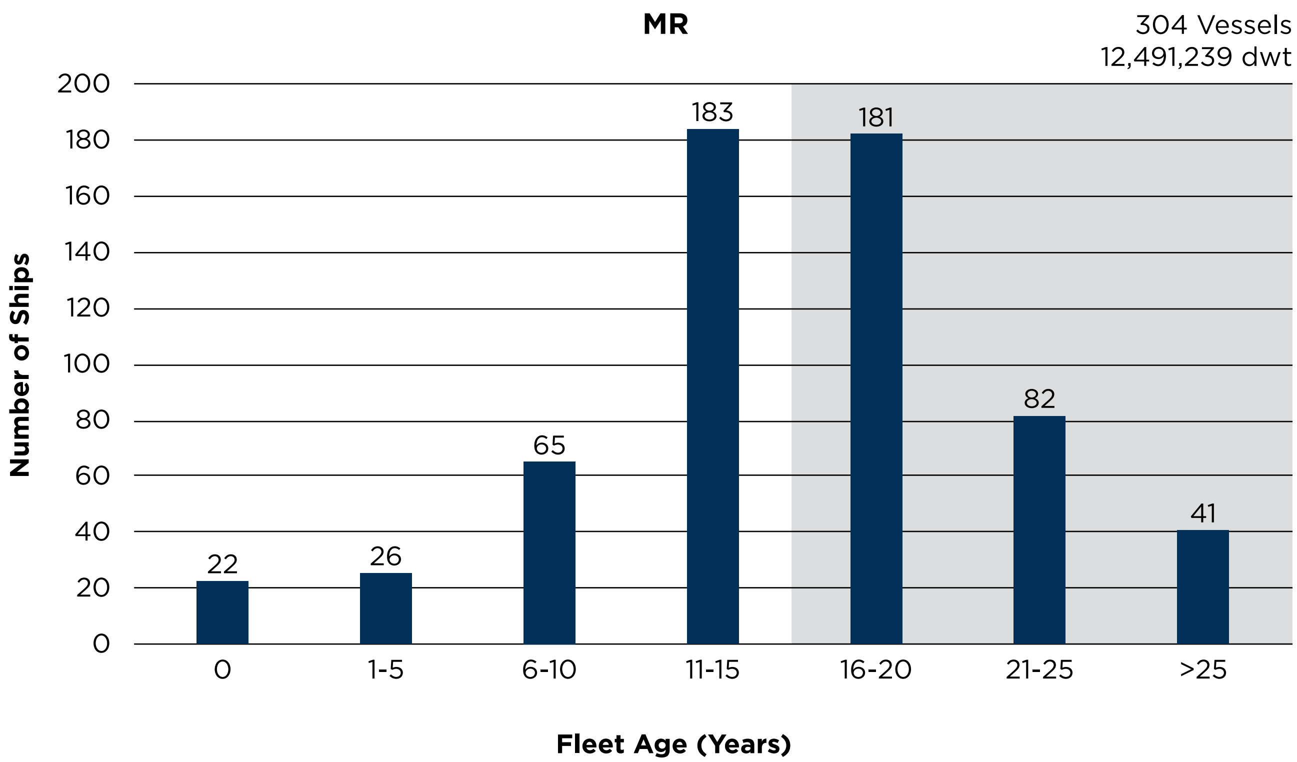


Figure 16: More than half of the global MR fleet is above 16 years old.

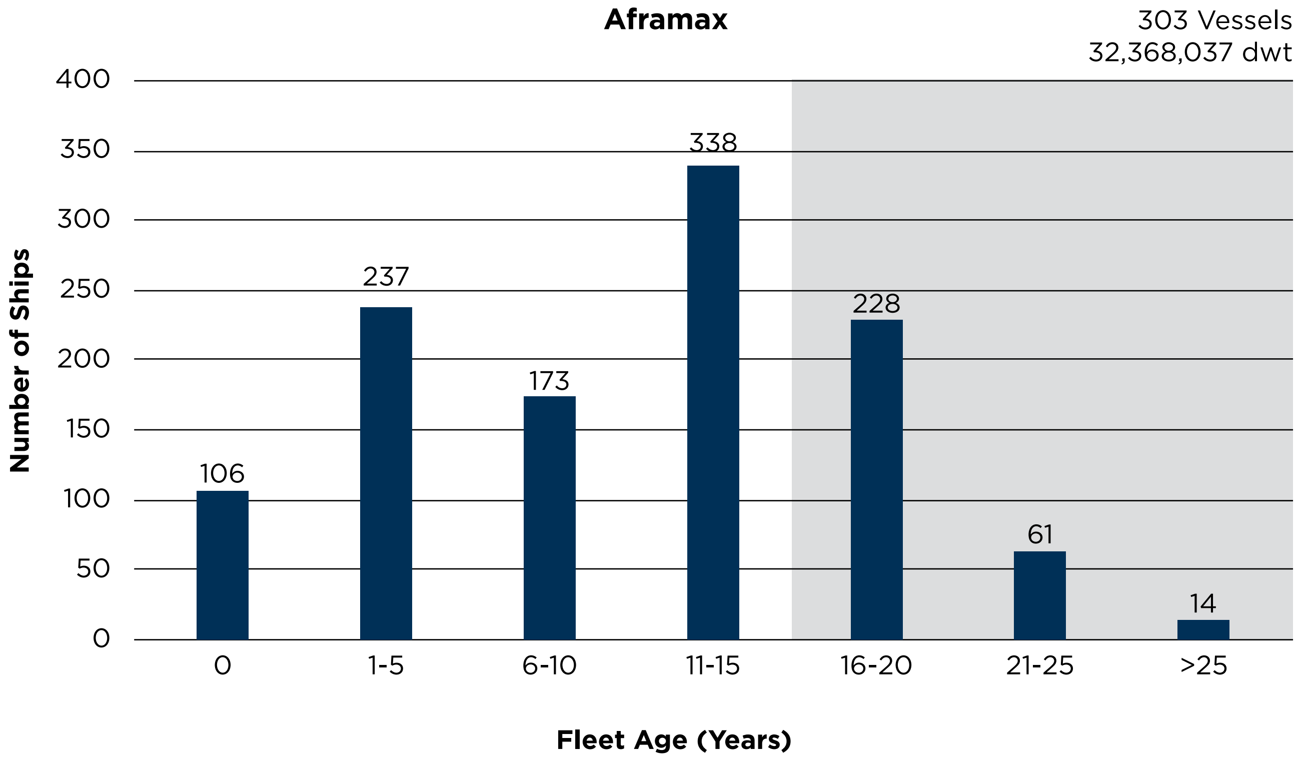


Figure 17: The largest proportion of the aframax fleet is 11 to 15 years old.

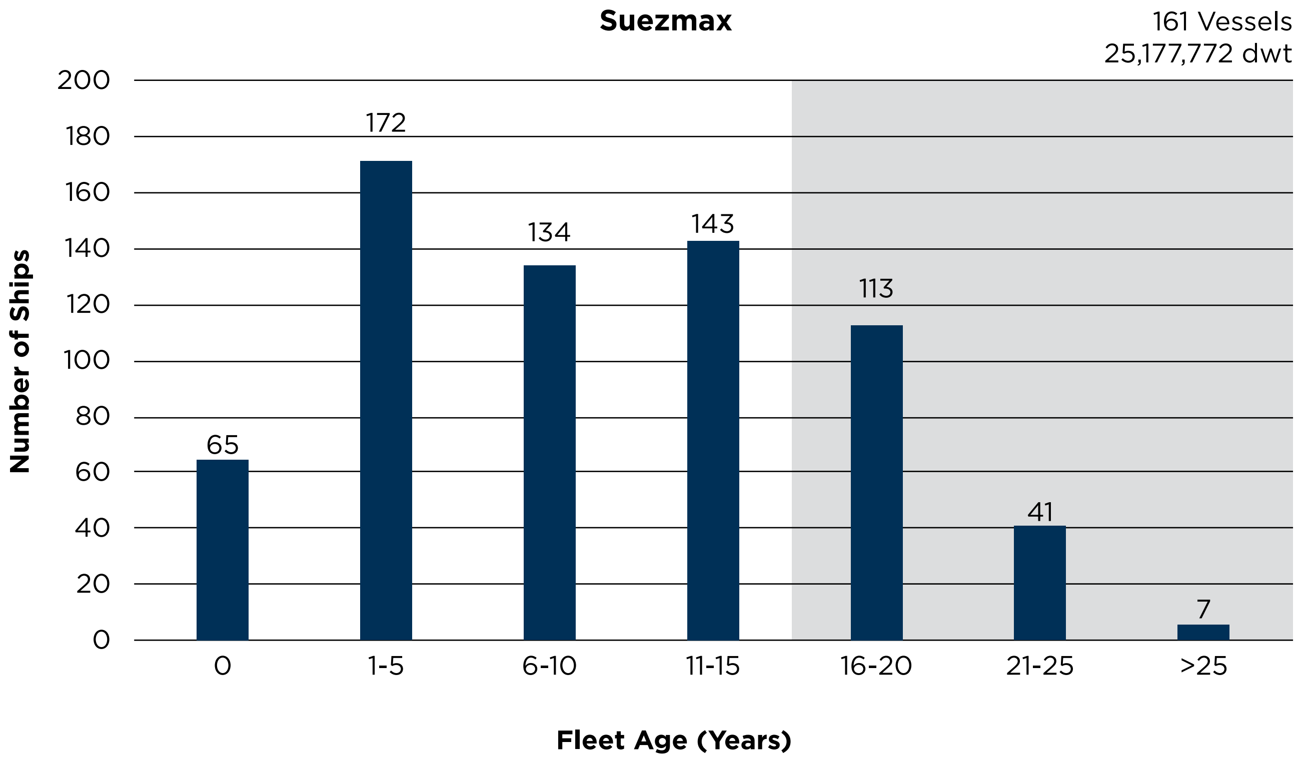


Figure 18: The suezmax fleet is the youngest of the three sectors examined.

Just over half of the in-service and on-order MR tanker fleet (about 51 percent of vessels and 48 percent of dwt) were 16 years or older. In general, the aframax (26 and 25 percent) and suezmax (24 and 24 percent) fleets were significantly younger.

TRADING ROUTES

The service trading areas for the MR, aframax and suezmax tankers are very different due to their nature and size. The charts below illustrate the operating routes and trading areas for the segments, based on AIS data records since 2018.

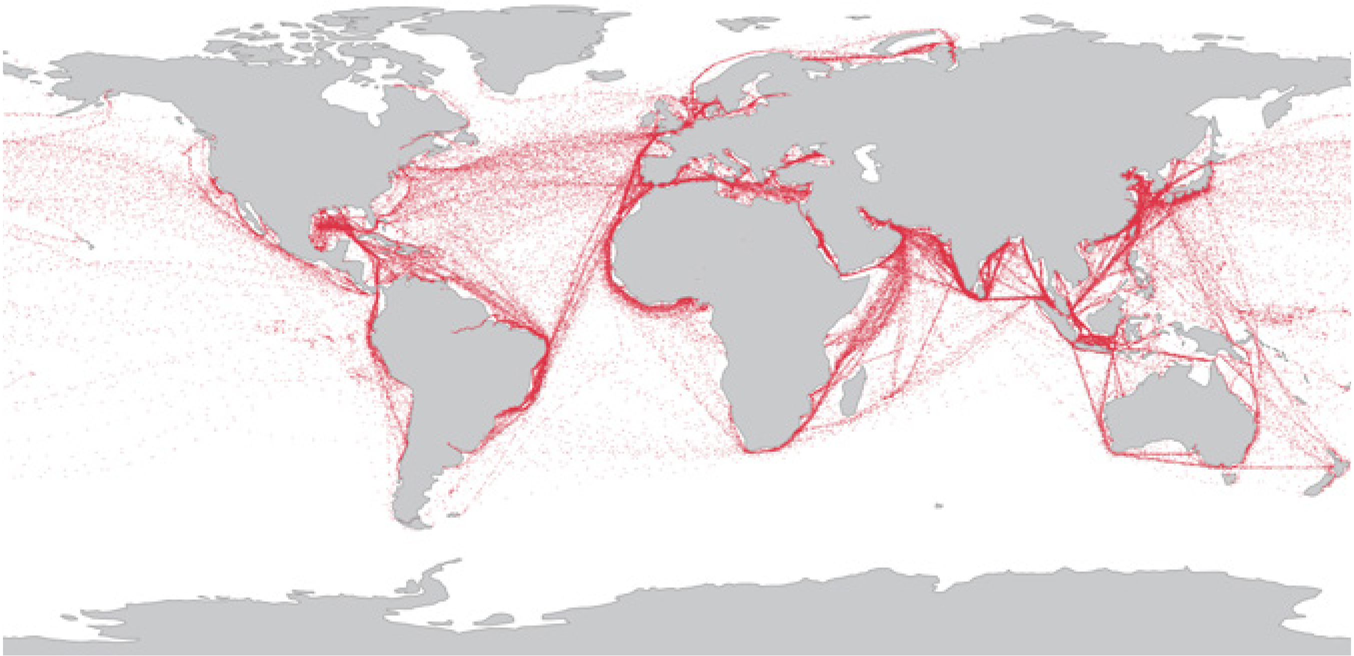


Figure 19: Distribution of MR tanker trading routes from 2018-2021.

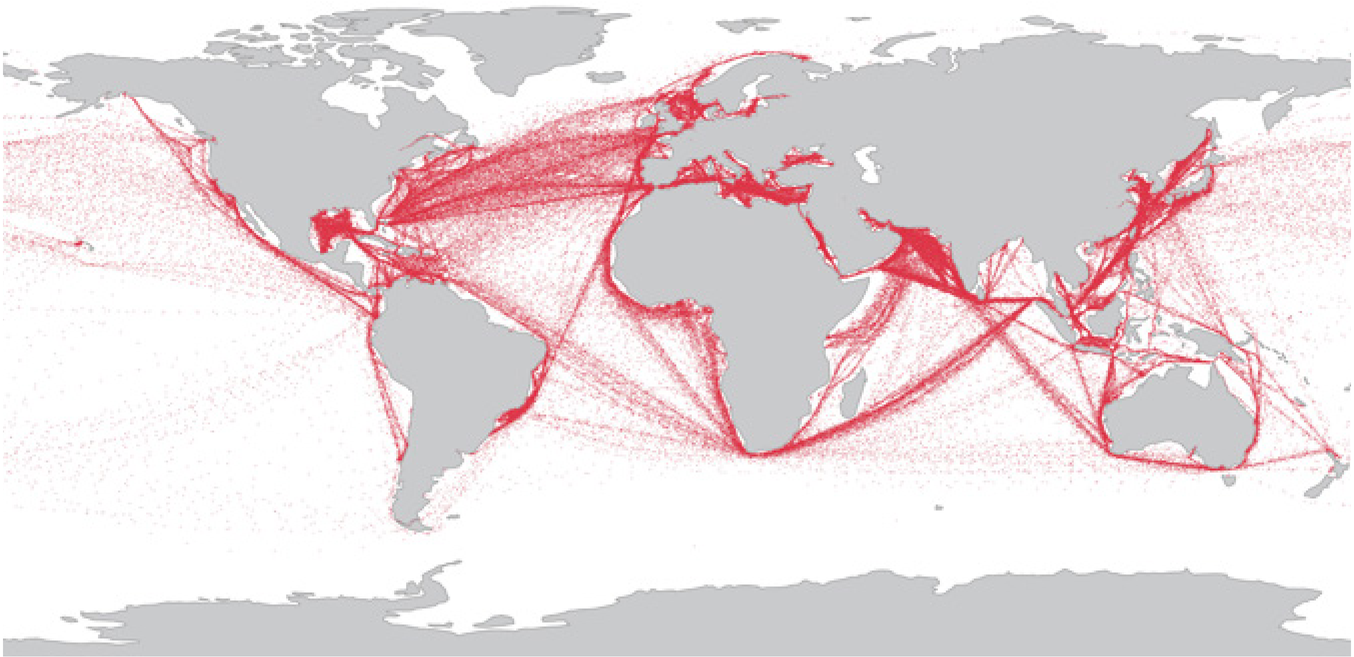


Figure 20: Distribution of aframax tanker trading routes from 2018–2021.

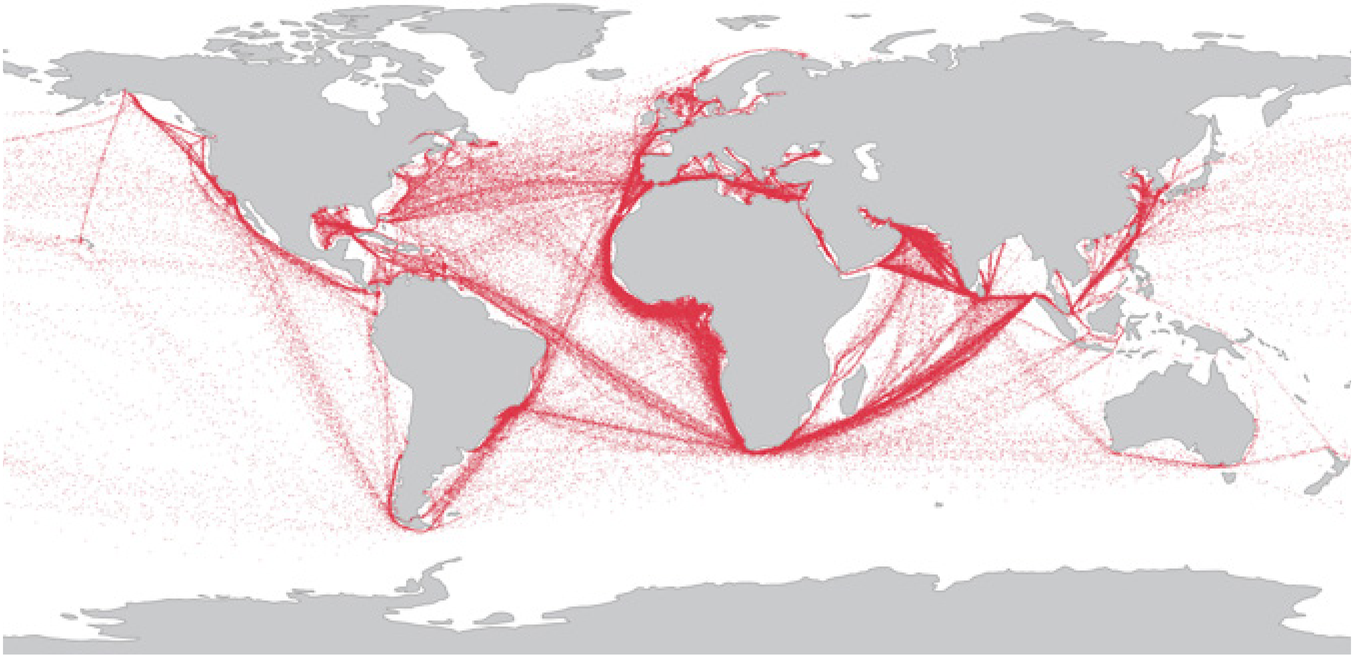


Figure 21: Distribution of suezmax tanker trading routes from 2018-2021.

TRENDS IN VESSEL SPEED AND DRAUGHT

The average operational speed and draught for these three segments remained relatively stable from 2018 to 2021, according to AIS data. The charts on this page and the next illustrate their speed and draught trends from 2018 to 2021.

For MR tankers, 45 percent of the fleet's operating speed was between 10.5 and 13.5 knots. Laden draught was typically 10 to 12 meters, while the ballast draught was from seven to nine meters.

For aframax tankers, 58 percent of the fleet's operational speed was between 10.5 and 13.5 knots. Laden draughts were recorded in a broader range, but typically from 11 to 15 meters, while the ballast draught was mainly between eight and nine meters.

The suezmax fleet also had a very narrow range of speed for laden and ballast voyages during the period; 61 percent of operational speeds were between 10.5 and 13.5 knots. Laden draught was typically around 15 to 16 meters, while ballast draught was nine to 10 meters.

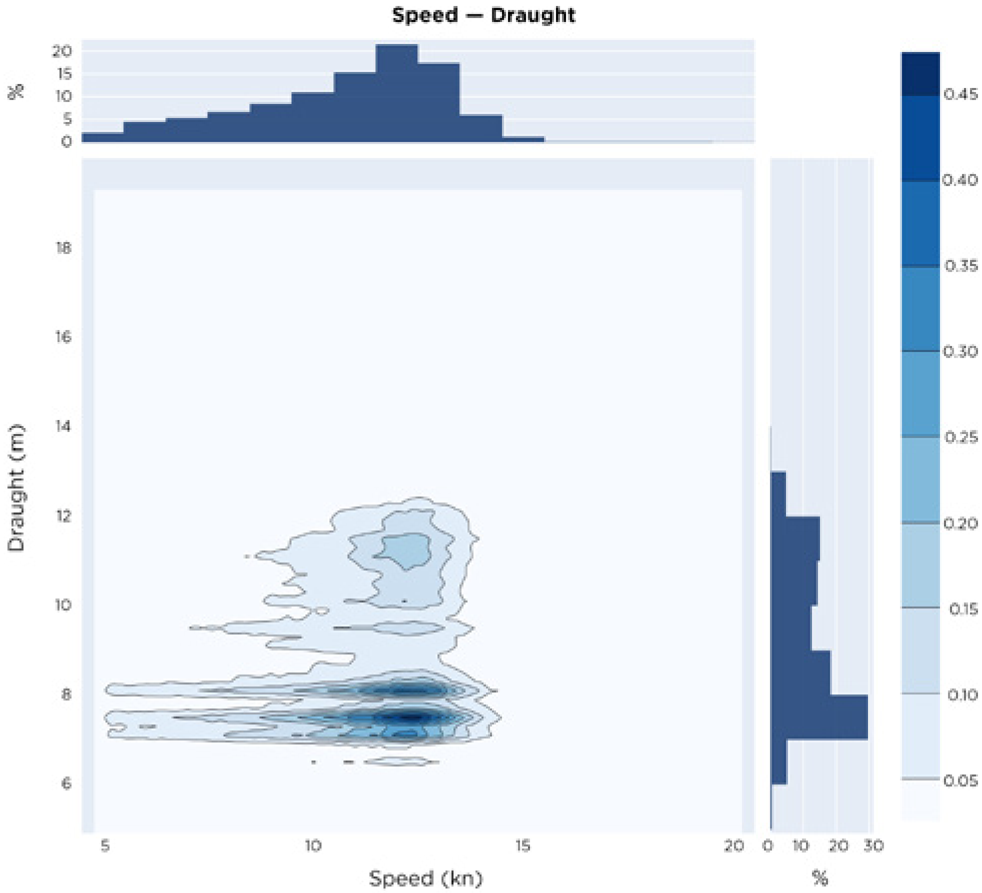


Figure 22: MR tanker speed-draught chart.

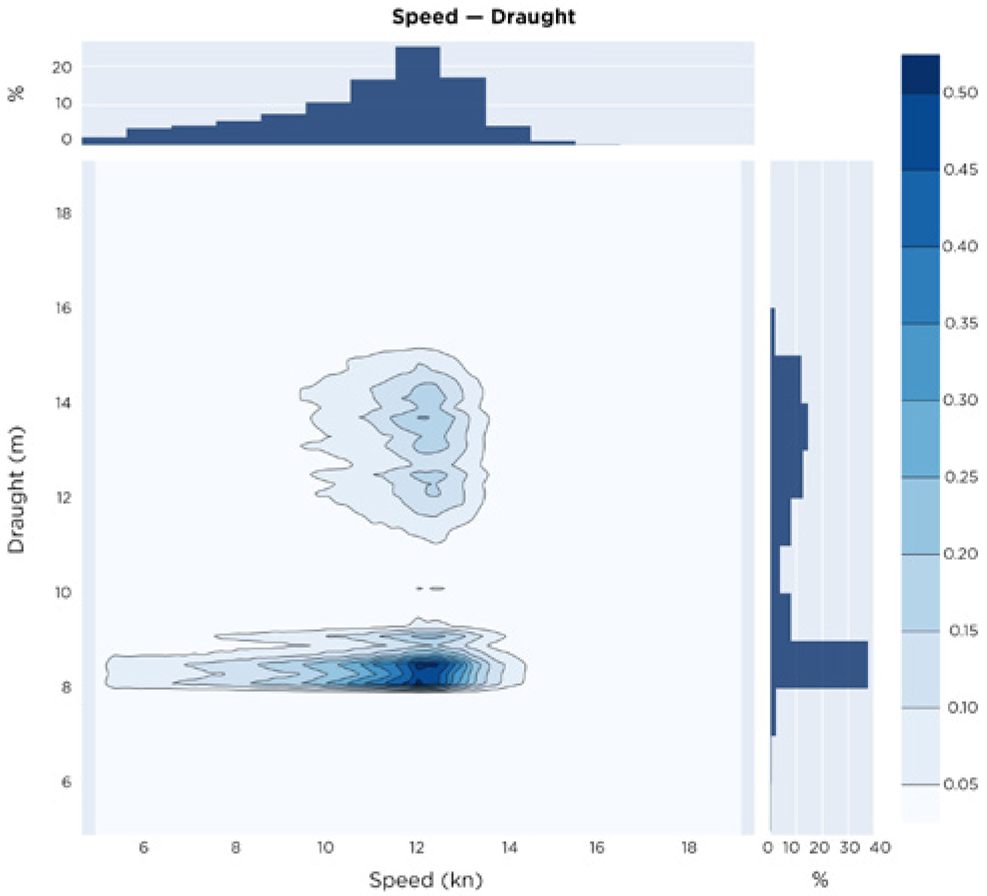


Figure 23: Aframax tanker speed-draught chart.

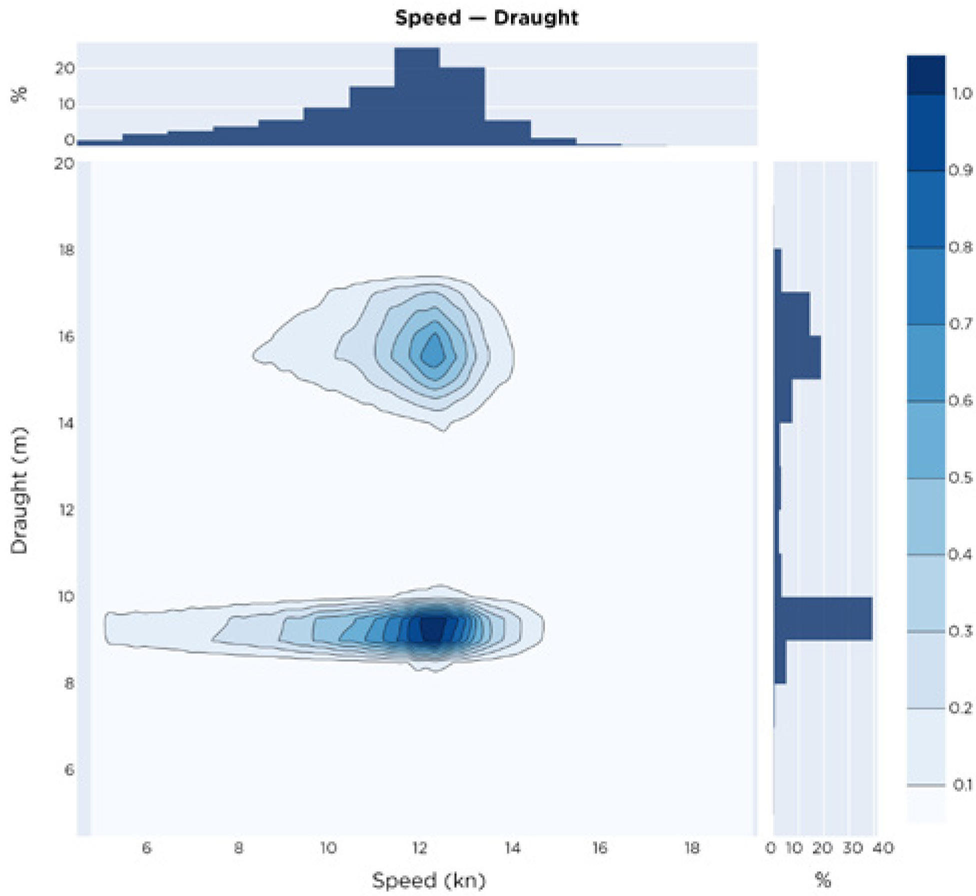


Figure 24: Suezmax tanker speed-draught chart.

ENERGY EFFICIENCY OPTIONS

All of the tanker segments being reviewed were predominantly equipped with conventional propulsion systems, e.g., two-stroke, slow-speed engines driven by fixed-pitch propellers.

The majority of the propulsion engines were diesel-fueled, using liquid forms. A small number of aframax and suezmax tankers (45 aframax and eight suezmax, according to Clarksons data) had installed dual-fuel (DF) engines to burn liquefied natural gas (LNG).

For the smaller MR tankers, a few vessels (nine) had adopted diesel-electric propulsion. Batteries, serving as part of hybrid-power systems, were installed on four suezmax and two aframax tankers.

The following table summarizes some aspects of the installed fleet machinery, hull forms, energy-saving devices, etc., according to analysis of recent construction trends.

	ITEM	MR TANKERS	AFRAMAX	SUEZMAX
Hull	Form	Hull form optimized to improve efficiency; finer aft hull	Hull form optimized to improve efficiency; finer aft hull	Hull form optimized to improve efficiency; finer aft hull; fuller forward hull form with straight (bulb-less) stem profile
	Steel	Increased use of higher strength steels to reduce weight	Increased use of higher strength steels to reduce weight	Increased use of higher strength steels to reduce weight
	Arrangements	Larger spacings for web frames	-	Smaller forepeak tank to reduce the moments that bend design
	Rudder	Semi-spade rudder	Full spade with twisted leading edge and bulb appendage	Full spade with twisted leading edge and bulb appendage
	Energy-Saving Devices	-	Boss-mounted half duct	Boss-mounted half duct
Machinery	Propulsion Engine	Dual-fuel engine for emission regulation, reduced power owing to improved hull form	Dual-fuel engine for emission regulation	Dual-fuel engine for emission regulation
	SO_x	SO_x scrubber or dual-fuel engine	SO_x scrubber or dual-fuel engine; hybrid SO_x scrubber	SO_x scrubber or dual-fuel engine; hybrid SO_x scrubber
	EEDI	Phase II and preparing for earlier Phase III	Phase II	Phase II

TECHNICAL MEASURES

Adopting a variety of technical measures has helped to improve the operating efficiency of tankers and reduce the associated costs. Some of the typical measures included the use of lighter weight materials, additional hull coatings, using air to lubricate the hulls, reducing ballast water, optimizing hull forms, adopting hybrid-power/propulsion-power systems, optimizing machinery, installing waste-heat recovery (WHR) systems, and wind-assisted propulsion systems, etc.

To date, the main technology adopted for larger tankers, particularly in the VLCCs segment, has been WHR, which is typically used to supplement onboard power requirements.

However, experimental case studies reviewed at MEPC 72 and after found that the adoption of energy-saving devices such as pre/post-swirls devices, contra-rotating propellers, low-friction coatings, WHR and solar power could also help to increase the design-based efficiency of the global tanker fleet.

SUMMARY OF ENVIRONMENTAL-RELATED NOTATIONS

ABS offers class notations to recognize efforts made by owners and operators with regard to environmental protection and pollution control. They cover air emissions, water discharge, ship recycling, noise, oil pollution and overall environmental compliance. The notations are summarized in the following table.

AIR EMISSION	**EGC-SO_x**	Exhaust Gas Cleaning for SO_x	ABS *Guide for Exhaust Emission Abatement*
	EGC-SCR	Exhaust Gas Cleaning – Selective Catalytic Reduction (for NO_x)	
	EGC-EGR	Exhaust Gas Cleaning – Selective Catalytic Reduction (for NO_x)	
	NO_x Tier III	Vessels with marine diesel engines meeting IMO NO_x Tier III	
	EEMS	Exhaust Emission Monitoring System (for SO_x or NO_x)	
	SO_x Scrubber Ready	SO_x Scrubber Ready – Level 1, Level 2 and Level 3	ABS *Guide for SO_x Scrubber Ready Vessels*
AIR EMISSION (CARGO VAPOR)	**VEC, VEC-L**	Vapor Emission Control	ABS *Marine Vessel Rules*
WATER DISCHARGE	**BWE**	Ballast Water Exchange	ABS *Guide for Ballast Water Exchange*
	BWT, BWT+	Ballast Water Treatment	ABS *Guide for Ballast Water Treatment*
SHIP RECYCLE	**IHM**	Inventory of Hazardous Material	ABS *Guide for the Inventory of Hazardous Materials*
NOISE	**UWN, UWN+**	Underwater Noise Notation	ABS *Guide for the Classification Notation Underwater Noise and External Airborne Noise*
	AIRN, AIRN+, AIRN-M	External Airborne Noise	
OIL POLLUTION	**POT**	Protection of Fuel and Lubricating Oil Tanks	ABS *Marine Vessel Rules*
OVERALL ENVIRONMENT	**ENVIRO, ENVIRO+**	Environmental Protection for vessels	ABS *Guide for the Environmental Protection Notation for Vessels*
SUSTAINABILITY	**SUSTAIN-1 (2020) SUSTAIN-2 (2020)**	Demonstrate the sustainable development goals related to vessel design, outfitting and layout that can be controlled, measured and assessed	ABS *Guide for Sustainability Notations*

JOINT DEVELOPMENT PROJECTS (JDP)

ABS has been working with industry partners to improve the sustainability performances of tankers by improving their energy efficiency, minimizing air emissions and reducing the ships' carbon footprints.

Some of these initiatives have been realized through partnership frameworks such as JDPs, formed to investigate opportunities such as the ship-specific viability of using DF engines, alternative fuels, hybrid-power systems and fuel cells for propulsion.

The JDPs also have been used to study the application of digital technologies. A transition to enhanced digitalization can help owners to address the challenges of decarbonization. However, increasing onboard connectivity also can introduce cyber-related risks that would need to be addressed.

Partnership vehicles such as JDPs encourage industry to develop practical solutions and simultaneously create a framework to address the decarbonization challenges without compromising cyber security defenses.

Below are some of the key JDPs involving shipyards, owners and equipment makers:

- With shipyard for LNG dual-fueled tanker
- With shipyard, owner and charterer on LNG's application for aframax tankers
- With shipyard, owner and engine manufacturer for ammonia-fueled tanker
- With shipyard, owner and charterer for LNG-fueled hybrid tanker
- With shipyard for decarbonization and digitalization
- With shipyard for carbon-footprint assessment by simulating the application of technology measures, alternative fuels and operational scenarios
- With shipyard and owner for implementing fuel cell as main propulsion for tanker

These are just a few of the industry projects recently initiated specifically to explore the viability of solutions that have the potential to lower the carbon footprint of tankers and promote their longer term sustainability and energy efficiency.

KEY DESIGN FEATURES FOR THE NEXT GENERATION OF TANKERS

ACHIEVING EEDI PHASE III COMPLIANCE

Changes in tanker designs – both for existing and future ships – are imminent. The regulatory drivers for the changes include the introduction of the CII – the IMO's new short-term operational measure to reduce the carbon intensity of international shipping by at least 40 percent by 2030 (compared to 2008) – the EEDI requirements (for newbuilds) and the EEXI technical measures for existing ships.

Discussions are ongoing about introducing medium- and long-term measures to further reduce GHG emissions, including MBMs (such as carbon levies, emissions caps and carbon-trading schemes) and introducing new ways to measure the full life cycle, well-to-wake (WtW) GHG emissions from conventional fuels and the alternative energy sources that ultimately may power ships.

Unfortunately, there is no "silver bullet" for achieving compliance with these regulations and targets, nor are there well-defined strategies on how a specific vessel or fleet could plot efficient courses to low- and zero-carbon shipping. The global tanker fleet consists of 10,309 vessels of more than 4,000 dwt, out of which 1,903 have attained EEDI values, or about 18.5 percent of the global fleet that qualified. Some 77.3 percent of those tankers (or 7,974 ships) have yet to attain EEDI values; the remaining 432 vessels are not subject to the mandatory submission of EEDI data.

Table 1 shows the EEDI compliance levels for the tankers with attained EEDI values, according to the IMO GISIS database. While the IMO is constantly strengthening the EEDI targets by increasing the rates at which emissions must be reduced and bringing forward the programs' start dates, it is obvious that compliance with EEDI Phase III – and in some cases even Phase II – will remain a challenge for the majority of the fleet.

ATTAINED EEDI AVAILABLE - 1903	VESSELS	TOTAL DEADWEIGHT
EEDI Pre - Phase 0	2	637,514
	0.1%	0.3%
EEDI Phase 0 ~ Phase 1	35	4,991,302
	1.8%	2.6%
EEDI Phase 1 ~ Phase 2	494	91,030,738
	26.0%	47.8%
EEDI Phase 2 ~ Phase 3	798	69,880,806
	41.9%	36.7%
> EEDI Phase 3	574	23,837,996
	30.2%	12.5%

Table 1: EEDI compliance levels for tankers with attained EEDI values.

Because this emerging regulatory framework is significantly demanding, the next generation of tanker designs is expected to evolve and be adapted in a market where emission and sustainability profiles will be among the key selection criteria for vessel chartering. The transition will affect two general areas: fuel selection and powering systems, and hull designs and construction.

The IMO's GHG strategy for decarbonization – and the European Union's (EU) goal to be climate neutral by 2050 – make the transition to alternative fuels (either low-carbon or carbon-neutral) unavoidable; in fact, switching to "greener" fuels is widely seen as the most effective way to meet the new regulatory metrics (EEDI, EEXI, CII). That said, this does not mean that the potential for improvements in architectural design, energy-efficient technologies and operational measures to affect the rate of decarbonization should be overlooked.

FUEL SELECTION AND POWERING SYSTEMS

The most viable way for a newbuild tanker to achieve compliance with EEDI Phase III and prepare to comply with any other medium- and long-term measure is to adopt low- and zero-carbon fuels.

As can be seen from the graphic on the following pages, LNG (and methane gas) and methanol (industry data soon will be further enriched by the 40 liquefied petroleum gas [LPG] main engines on order) are the most mature fuels in terms of available infrastructure and technological readiness. Reliable engine solutions, fuel supplies and the infrastructure for handling and storage are already available.

The current trend in the tanker newbuildings scene features mostly aframax-sized tankers powered by low- or high-pressure dual-fuel (DF) engines (XDF – ME-GI) coupled usually with two Type C tanks with a fuel capacity of about 900 cubic meters (m^3) each. The Type B self-supporting prismatic tank is a less popular option, usually coupled with high-pressure DF engine on VLCC type tankers, having a fuel capacity of about 1,600 to 3,000 m^3. There also have been significant developments in the LNG DF engine market, as engine makers have come up with innovative designs and systems to reduce methane slip, improve combustion stability and control nitrogen oxide (NO_x) emissions. There is also a continuous improvement in fuel gas supply systems as new safer and more efficient designs have emerged offering variable natural boil-off gas (BOG) handling configurations.

The use of methanol DF engines is also a very popular option particularly for small to medium-sized methanol carriers. However, there have been some proposed larger tanker designs that consider two Type C tanks with a volume of about 9,000 m^3, but the increased fuel storage space and operational expenditures (opex) compared to LNG make it a less appealing configuration.

The combination of these reasons makes LNG and methanol the best interim alternative fuel solutions until other technologies advance to support safe and commercially sustainable vessel powering from zero-carbon fuels such as hydrogen and ammonia.

For the time being, the challenges associated with ammonia's toxicity, corrosiveness and relatively low energy density — as well as hydrogen's very low fuel density in a liquid state — have not been addressed to fully support their frequent use in the maritime sector. But an ammonia engine is expected to be commercially available by the end of 2024, and "ammonia ready" designs are starting to emerge. The first hydrogen engine already has been introduced to the market. These designs will use DF engines that could run on LPG with arrangements (such as reserved space for an ammonia tank, a dedicated fuel-supply system for ammonia in a liquid state and a fuel-preparation room) to support the transition to using ammonia at a later stage. The fuel-gas supply system and engine would need to be converted; the tanks, however, could be reused.

Volatile Organic Compounds (VOC) Recovery Systems are also coming to light. These systems which can be coupled with liquid gas (methanol or LPG) DF engines, utilize the light components of crude oil that evaporate mainly during loading operations. After the reliquefaction of this surplus gas, VOCs are used as either main engine fuel or reinjected back into the cargo tanks.

Additionally, the use of bio- and synthetic fuels, such as Fatty Acid Methyl Ester (FAME), biodiesel, hydrotreated vegetable oil, dimethyl ether and other options deserve consideration, as they can be used in existing engines and would require no additional or dedicated bunkering infrastructure. However, the carbon life cycle impact of those fuels has yet to be fully measured.

The matrix set out below presents the key properties of some alternative fuels, their indicative costs, applicable levels of marine-readiness, feasibility levels for implementation, tank-to-wake (TtW) and WtW emissions, and indicative capital expenditure (capex) for the associated engines. It offers insights into the advantages and disadvantages of their use in existing or newbuild tanker designs.

Fuel Alternatives	Energy Density (MJ/L)	Liquid Density (kg/m³)	Boiling Point, ° C 1 bar	Conventional or Cryogenic tanks	Required Tank Volume m³	Ignition Energy	Bunkering Availability – Infrastructure Readiness	Implementation Feasibility – Retrofit	Implementation Feasibility – Newbuilding	Technology Readiness Level (TRL 1-9)	Engine Technology Capex
LNG (Light Gas and Alcohol Pathway)	22.2	450	-162	CRYO	1,590	LOW-MEDIUM	Sufficient level of Bunkering Readiness - Growing The extensive LNG bunkering infrastructure and the growing number of LNG bunker barges/ vessels in combination with the decent energy density of the fuel make it the most appealing choice for the time being	HARD	EASY	9	HIGH Low Pressure LNG DF Main Engine Capex (approx. $300/kW) High Pressure LNG DF Main Engine Capex (approx. $350/kW) LNG DF GenSets (approx. $420/kW) + Pressurised Tanks + FSS system + EGR/SCR + Nitrogen Supply + Fuel Valve Train + Reliq/ Subcooling System etc.
Hydrogen (Light Gas and Alcohol Pathway)	8.5	76.9	-253	CRYO	4,117	LOW	Insufficient Level of Bunkering Readiness Limited number of bunkering vessels available at the moment. No existing maritime handling regulations	HARD	HARD	3	HIGH Internal combustion engine (ICE) options are not commercially appealing due to the low liquid density of the fuel. An estimated capex would be too high
Ammonia (Heavy Gas and Alcohol Pathway)	12.7	600	-33	CONV	2,755	HIGH	Insufficient level of Bunkering Readiness There is an extensive ammonia distribution network already in place, but not suitable for bunkering purposes; however, there is strong long-term scalability potential	HARD	MODERATE	5	HIGH Ammonia DF Main Engine Capex (approx. $400/kW) + Appropriate Gen Set (approx. $500/kW) +Pressurised Tanks + FSS System + SCR aftertreatment system
Methanol (Heavy Gas and Alcohol Pathway)	15.7	798	65	CONV	2,333	LOW	Insufficient level of Bunkering Readiness Limited number of bunkering vessels available at the moment. No existing maritime handling regulations for the time being; however, existing infrastructure can be easily be repurposed	MODERATE	EASY	9	HIGH Methanol DF Main Engine Capex (approx. $400/kW) Methanol FSS (approx. $40/kW) + Fuel Service Tanks + Appropriate Gen Set (approx. $500/kW) + SCR aftertreatment system
LPG (Propane) (Heavy Gas and Alcohol Pathway)	26.5	492	-42	CONV	1,346	LOW-MEDIUM	Sufficient level of Bunkering Readiness	MODERATE	EASY	8	HIGH LPG DF Main Engine Capex (approx. $350/kW) + Appropriate Gen Set (approx. $500/kW) + Fuel Service Tanks + Low Flashpoint Supply System + SCR aftertreatment system
Biofuels *Indicative Values Based on Biodiesel E30	33-40	991	315-350	CONV	1,100	LOW-MEDIUM	No need for new infrastructure/ bunkering facilities	EASY	EASY	8	HIGH ICE engines are commercially available with mature Capex (approx. $275/kW) The investment is mostly affected by the price of the biofuels and future regulations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Net power savings and EEDI figures can be improved by applying any number of existing or innovative energy-producing technologies, most of which hold considerable potential to support shipping's transition to a zero-carbon future.

For example, a shaft generator/power take off option can provide a reliable source of green energy, with the generator placed in the shaft line between the two-stroke engine and the propeller – or coupled directly to the fore end of the main engine – to harvest the kinetic energy produced by the engine to produce electricity.

Depending on the configuration, the electricity can be either stored in a battery system, used for the "hotel load," or to power a thruster and other secondary motors, enhancing the operational efficiency of the tanker (ideal for DP shuttle tankers) and lowering its carbon footprint.

Onshore power supply facilities (such as switchgears, transformers, cables, etc.) can be installed at relatively moderate costs in tankers' engine rooms and, subject to port availability, allow the vessels to be supplied with "green" electricity (through processes such as "cold ironing") during loading and unloading operations.

Battery systems or fuel-cell applications are another consideration. Despite their well-known cost and space limitations, the transition to battery technologies has gained considerable momentum due to their accelerating pace of adoption in the automotive sector. However, hybrid-battery solutions really only should be considered as a solution for tankers whose operations require high peak loads; all-electric lithium-ion battery designs are currently best suited for smaller, coastal tankers.

Similarly, the adoption of fuel cell technology for marine use (solid-oxide, proton exchange membrane fuel cells, etc.) is slowly advancing. However, for the next 10 to 15 years, their use is expected to be limited to "complementary" technologies in hybrid tanker configurations, not as options for primary propulsion.

Lastly, post-combustion, carbon-capture technologies that clean exhaust gas are expected to find a place in the next generation of tanker designs. Carbon-capture systems could include technologies that use techniques such as liquid absorption, membrane separation, etc., with a carbon-treatment and storage systems located at the aft of the vessel behind the funnel area.

HULL DESIGN AND CONSTRUCTION

"Efficiencies-of-scale" concepts are expected to have a significant impact on future designs, simply because larger tankers can transport more cargo at a speed that requires less power per unit of cargo volume, and thus produce less grams of CO_2 per ton-mile.

By making ship designs more slender, increasing ship lengths and lowering the block coefficient Cb, (to a value close to eight), wave-making resistances will be reduced along with ships' overall fuel consumption. The length-to-breadth ratios for tankers may need to be redefined to fit the needs of a more sustainable market. While the effects of higher construction costs, ship weights and wetted surface areas should be factored into finding the optimum designs.

Additionally, the shape of the tankers' bulbous bows can be parametrically optimized to match their operating profile (considering variables such as optimum speeds, wind and loading conditions) to minimize wave making-resistance and the required propulsive power.

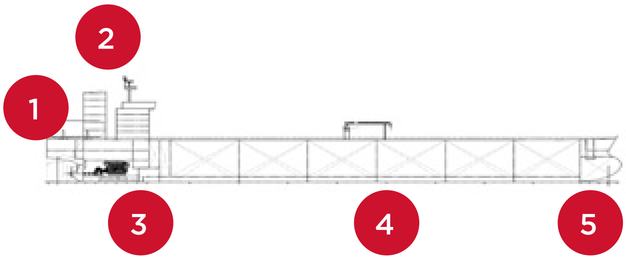
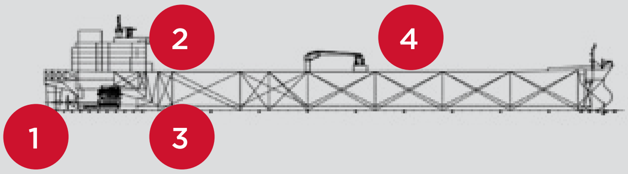
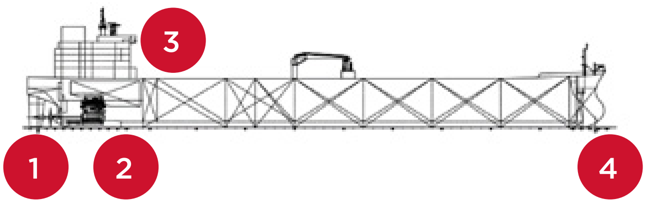
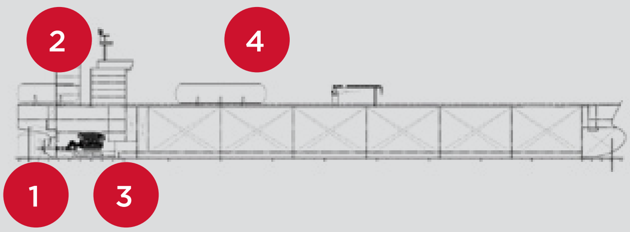
Energy-efficient technologies, such as propeller and rudder energy-saving devices (see the chart on the following pages) and air lubrication systems (ALS), will support the better hull forms and the hydrodynamics of design. ALS minimize frictional resistance by releasing constant streams of air bubbles from several openings along the hulls' flat bottom areas. But if future tanker designs offer a V-shaped hull to minimize the needs of ballast, this solution might be rendered obsolete.

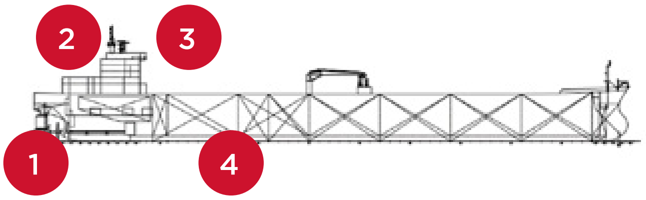
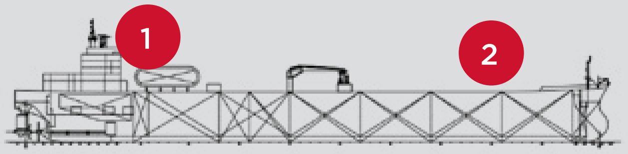
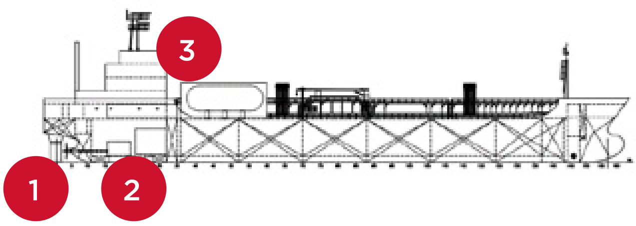
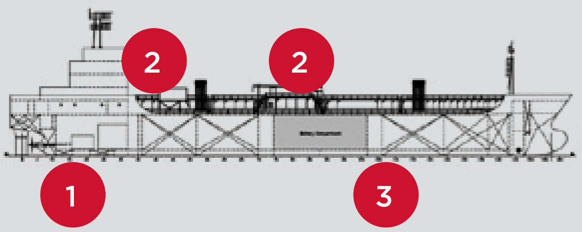
Other factors, such as minimizing hull openings (sea chests, bow thruster tunnels), shaft-line arrangements and skeg shapes are also considerations for the next generation of tanker designs.

Aerodynamic improvements can be made by streamlining the shape of the accommodation area or by installing windshield structures that reduce air resistance and lower fuel consumption. Wind-assisted ship propulsion (WASP) will play an important role in optimizing future designs, as wind rotors, sails and kites can convert kinetic energy into a thrust surplus that will minimize a tanker's powering requirements. Depending on the WASP configuration (number, location and dimensions of sails, operating profile, etc.), varying levels of power savings are possible.

Finally, future tanker designs are expected to be significantly lighter simply because removing unnecessary weight from the ship's structure, lighter displacement and lower resistance decreases their demand for power. This can be achieved by minimizing the use of ballast, with the ultimate goal being a "ballast-free" design.

The greater use of lightweight materials, such as high tensile steel for the vessel's outer shell or using aluminum in areas that do not directly affect a ship's overall strength, and composites can be expected, provided the development of new technologies and cost considerations do not eliminate their usefulness.

VESSEL DESIGN	KEY ASPECTS	TRL
Aframax Tanker — Dual Fuel very low sulfur fuel oil (VLSFO)/LNG  **Optimal Speed**: 13 knots **Range**: Approximately 8,967 nm **Fuel Consumption**: 35.2 MT/day (VLSFO) 78.3 m³/day (Gas)	1. 2 Type C pressurized tanks that can feed BOG with or without the need of compressors or high pressure pump systems 2. Gas supply room containing heat exchanger, LNG vaporizer units, master gas valves, BOG compressor, etc. 3. High- or low-pressure DF engine (tier II or III) depending on the fuel mode — coupled with min 3 DF gen sets 4. Air lubrication option 5. State of the art hull — propeller optimization based on the operating profile of the vessel	9
Liquid Biofuel (Fame) Suezmax Tanker  **Range**: Approximately 9,000 nm	1. Propeller design based on a derated main engine (ME), with minimal aerodynamic fairing 2. Conventional fuel storage, with slightly increased capacity because most biofuels are less energy dense than fossil fuels 3. Conventional internal combustion engine (ICE) and auxiliaries — ICE on biofuels have similar efficiencies to ICE operated on fossil fuels but have 7-8% lower heating value and are 10% lighter in mass density than HFO 4. Wind-assisted ship propulsion (WASP) option can be also considered	8
"Purely" LNG ICE Suezmax Tanker  **Range**: Approximately 9,000 nm	1. Optimized propeller design — PTO 2. Four-stroke purely LNG burning (spark plug ignition) engine (Otto Cycle) which eliminates CO_2 emissions 3. Optimized capacity LNG tanks (Type 3 or 2 prismatic or membrane) 4. Optimum hull design — bulbous bow optimization	7
Aframax Tanker — LNG + 20% LH_2 Fueled  **Range**: Approximately 14,587 nm **Fuel Consumption**: 48.2 m³/day (LNG) 76.1 m³/day (LH_2)	1. Fixed pitch propeller (FPP) — Power take-off (PTO) option 2. 2,250 m³ LNG Type C tank 3. DF Engine concept that can support liquid hydrogen and LNG burning 4. 3,548 m³ LH_2 Type C tank	4

VESSEL DESIGN	KEY ASPECTS	TRL
NH_3 Solid Oxide Fuel Cell (SOFC) Suezmax Tanker  **Range**: Approximately 9,000 nm	1. SOFC technology can result to an overall propulsive efficiency of around 60% compared to ICE — Contra-rotating propellers — Conventional shaft driven by 8.5 MW electric motor and a 5.4 MW Steerable pod 2. Fully electric concept design with all the necessary propulsive power provided by NH_3 powered SOFCs 3. NH_3 stored in 2 prismatic Type B tanks P/S in the ER — Option for Type C deck tanks 4. Fuel cells are sized to meet the maximum power capacity of 1.0 MW for auxiliaries — 13.9 MW for propulsion and a minimum battery capacity of about 169 MWh used for power conditioning, dynamic energy stability and maintaining speed	4
Hydrogen Proton Exchange Membrane (PEM) Fuel Cell Suezmax Tanker  **Range**: Approximately 9,000 nm	1. Innovative containment system, reliquefaction plant, compressed gas storage, and battery rack systems are required 2. Similar Design to NH_3 SOFC suezmax Tanker; however, 8,820 m^3 tank capacity needed for liquid hydrogen versus 3,265 m^3 for NH_3 SOFC	3
Hybrid NH_3 SOFC/Batteries MR Tanker — Short Haul  **Range**: Approximately 4,200 nm	1. The propulsion power is provided to a single high-performance propeller coupled with a rudder bulb by a 3.0 MW electric motor 2. NH_3 Fuel cells are sized to meet the maximum power capacity of 3.5 MW 3. Minimum installed battery capacity of 15 MWh. Battery set is sufficient to provide enough power to the motor to take the 20% sea margin for 1.5 days taking into consideration high auxiliaries load and cargo heating	3
Fully Electric MR Tanker — Short Haul  **Range**: Approximately 720 nm	1. 15 MWh Lithium-ion (Li-ion) battery set 2. Battery sets are set to provide maximum power capacity of 1.0 MW for auxiliaries and 13.9 MW for propulsion 3. Li-On Batteries — 200 MWh of energy, which increases the lightshipweight from 2,850 MT to 3,850 MT plus stress concentration point and payload loss	2

GETTING AHEAD OF THE REGULATIONS AVALANCHE — WAYS TO IMPROVE EEXI, CII

The CII will be calculated using either an Average Efficiency Ratio (gCO_2/dwt-nm) based on the designed deadweight of the vessel, or through the Energy Efficiency Operational Index (gCO_2/ton-nm), which uses the cargo mass. The information will be sourced from the IMO DCS, the same system through which the CII will be calculated and reported.

Several of the CII's key variables – such as the exclusions and exceptions that could be applied for tanker operations (cargo heating/DP), environmental conditions (adverse weather/ice conditions, etc.) or voyage details (medium-sized tankers with frequent port calls) – have not been defined yet. This complicates the decision-making process for tanker shipowners when it comes to devising a CII strategy. However, after MEPC 78 (in the first half of 2022), most of these issues will be settled.

The two best short-term solutions for current tankers to be ranked within the A-C scale (see previous section on regulation) are to reduce speed and/or retrofit energy-saving devices, or a combination that optimizes results. Of course, any strategy that significantly reduces speed to lower fuel consumption and emissions output will need the support and consent of charterers.

Based on a recent analysis performed by ABS, larger tankers proved more able to change a CII rating by reducing speed. The CII is a ratio between CO_2 emissions and transport work (distance traveled multiplied by a vessel's deadweight or gross tonnage). So, when the speed is reduced, CO_2 emissions and the CII value are also reduced; at the same time, when the annual distance traveled in a year declines, the CII value increases.

As the transport work takes into account the deadweight and not the actual cargo transported, the effect of any reduction in speed is the same for all vessels, irrespective of their size. Therefore, the ability to reduce the CII rating relies mainly on the ability to reduce a vessel's CO_2 emissions.

The figure below shows CO_2 emissions (made nondimensional) as a function of speed. It illustrates that larger vessels are more able to use speed to change CO_2 emissions. The database used for the graphs below is for vessels of the same age; in principle, they should have the similar levels of design efficiency.

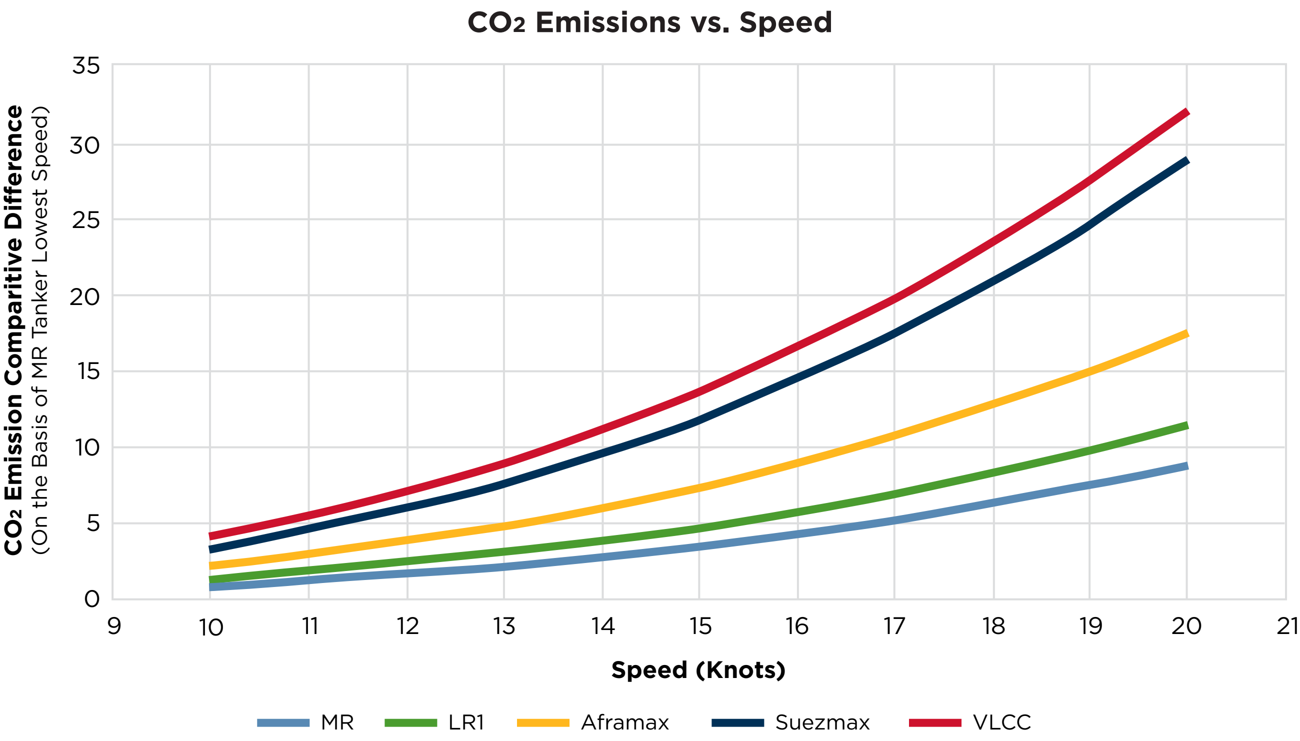


Figure 25: CO_2 emissions vs. speed.

OPERATIONAL MEASURES

Other options (which can be combined with those above) are operational procedures that can be identified by data analytics, such as real-time monitoring of ships' performances and fuel consumption, optimizing cleaning intervals for hulls and propellers, weather and trim optimization, using advanced hull coatings, "smart" (predictive) maintenance schemes and just-in-time (JIT) scheduling.

The adoption of JIT operations will reduce idling time while waiting for berths, minimize port stays, reduce fuel consumption and other port-associated costs. Moreover, it will substantially reduce the GHG and other ship-related emissions released while close to ports. A comprehensive list of the energy-efficient technologies that can be applied to a new design or retrofitted to an existing tanker, as well as the operational measures that can increase energy efficiency, is set out on the table on the following pages.

Energy Efficiency Technologies

	CLAIMED SAVINGS	COST — ESTIMATED CAPEX	ROI
Rudder Surf Bulb	5%	MODERATE	< 36 months
Rudder Surf Fins	1%	MODERATE	< 132 months
PBCF	1-3%	MODERATE	< 14 months
Contra Rotating Propeller (CRP)	3%	MODERATE	< 132 months
Mewis Duct	3-7%	MODERATE	< 14 months
Propeller Duct	3%	MODERATE	< 24 months
Wake Equalizing Ducts	2%	MODERATE	< 18 months
Pres-Swirl Fins	2%	MODERATE	< 30 months
Silicon Anti-Fouling Paints	6%	HIGH	< 9 months
Air Lubrication	4%	HIGH Estimated capex: ± $2 Million (M) - 3 M (estimate based on an aframax design)	< 60 months
De-rated ME	<6%	LOW	< 60 months
Part Load Optimization	3%	MODERATE	N/A
Turbocharger Cut-Out	3.5%	NO	N/A
Waste Heat Recovery Generator	2-5%	HIGH	< 72 months
Installation of LED Lighting	1-2%	HIGH	< 60 months
Solar Panels for Auxiliary Loads	5% of auxiliary Fuel	HIGH	< 60 months
Rotor and Wing Sails	10%	HIGH Rotor sail estimated capex: ± $1.5 M Wind sail estimated capex: ± $3.5 M (estimate based on 4 units systems)	< 60 months
Contracted and Loaded Tip Propeller (CLT) or Kappel Propellers	6%	MODERATE	< 12 months
Grim Vane Wheel	3%	MODERATE	< 60 months
PTO	Up to 35%	HIGH Estimated capex: ± $1.75 - 2 M (depending on engine manufacturer)	N/A
Onboard Carbon Capture System	N/A	HIGH	N/A
Batter Rack Hybrid System	10% of auxiliary Fuel	HIGH	N/A
Fuel Cells	N/A	HIGH	N/A

Operational Measures

	CLAIMED SAVINGS	COST — ESTIMATED CAPEX	ROI
Propeller Polishing	3%	LOW	< 6 months
Hull Cleaning	10%	LOW	< 1 month
Slow Steaming	20-36%	LOW	N/A
Virtual Port Arrival	6%	NO	N/A
Propulsion Efficiency Monitoring	5% over 5-year interval	MODERATE	< 24 months
Weather Routing/Software	5%	LOW	< 24 months
Port Turn-around Time	1%	NO	N/A
Optimization of Ballast and Trim	1-4%	LOW	< 24 months
Speed Optimization	5%	LOW	< 12 months
Autopilot Adjustment	1%	LOW	< 18 months
Optimized Voyage Planning	5%	LOW	< 12 months
Optimum Use of Fans and Pumps	1%	NO	N/A
Optimized Use of Bow Thruster	0.5%	NO	N/A
Efficient Control of HVAC	2%	NO	N/A
VFD for Pumps, Fans and Other Electrical Equipment	8%	LOW	< 12 months
Cargo Heating and Temperature Control Optimization	10% of cargo consumption	LOW	< 12 months
Optimized Machinery Maintenance	4%	MODERATE	< 24 months

The EEXI regulation, which is essentially an index for technical- or design-efficiency, estimates the grams of CO_2 a ship emits per capacity tonne-mile according to reference conditions specific to that ship. It is a function of the installed engine power (kW), the specific fuel consumption of the main and auxiliary engines, a carbon factor representing the conversion of fuel to CO_2, vessel capacity and vessel-reference speeds.

To comply with the regulation, the attained EEXI for a vessel must be less than or equal to the "required EEXI." The attained value can be significantly improved by limiting the power of the main engine, which is the most cost-effective solution to date, by incorporating one or multiple energy-efficient technologies which, depending on their nature, can improve the hydrodynamic profile of the vessel and result in net power savings (see above graphic), and by adjusting the primary fuel of the main engine.

Changing the main fuel on an existing vessel will involve a substantial investment and the completion of a life cycle cost analysis that takes into consideration elements such as the vessel's operational profile and expected life-cycle, capital expenditures for retrofits, operational expenses such as fuel costs, return on investment, etc.

Retrofits can have a strong positive effect on CII, if a low-carbon or a carbon-neutral fuel is selected.

With all these variables in play for shipowners, it seems that the only certainty is that tanker designs are destined to change.

CONSIDERATIONS FOR INDUSTRY

Attempting to transition a tanker fleet to low- and no-carbon futures is an incredibly complex process for owners. These initiatives need to start at the design and construction stages and be augmented at every subsequent step by real-time operational decisions that respond to inefficiencies as they arise.

The emerging capabilities of the Information Age will provide new levels of intelligence that challenge the shipowner to make many complex decisions and to consider many new options that were not available just a decade ago.

On the fuels front, the owner will need to make data-driven decisions during the ship's operation to reduce fuel oil consumption and improve its energy efficiency. Building in the ability to make accurate data-driven decisions needs to be addressed with the builder at the precontractual stages of a ship's technical specification.

The vessel is a complex network of energy systems; most are interrelated and influenced by operating conditions. As operational efficiency is key to successful transition, this requires not only a good understanding of a system's behavior, but also the ability to forecast a deterioration in its performance.

Generating accurate data is key, so any measurement sensors and instruments will need to be calibrated prior to sea trials, and the calibration records provided to owners.

Owners will need ship-specific information for an array of objectives, including:

- To establish speed-power-consumption reference values in variable conditions (draft, weather, etc.)
- To build an improvement plan for regulatory compliance
- An econometer (load diagram, specific fuel-oil consumption mesh) for optimum operation
- To assess hull-propeller degradation
- To assess engine-fuel degradation
- To optimize voyage planning
- Other objectives

Building the ability to take action during live operations (i.e., when the result of the action influences directly the vessel's energy output) will require owners to identify the key energy systems and the specific information needed to monitor their performance.

This information will not be restricted to operational data; design details also will be required in many cases and will need to be confirmed with the builder at an early stage.

For example, consider the most common energy-influencing system: propulsion energy. For this, information will need to be generated from model tests that are carried out with the final wake adapted and an as-built propeller for at least three drafts and not less than six speeds (preferably 10).

The scope of the testing also will need to cover the lowest possible speeds (lower than those selected above, which should be presented using computational fluid dynamics) for which the towing tank can deliver accurate results.

This process will include a full model-test report with towing resistance, self-propulsion and propeller open-water characteristics, including the load-variation factors KSIp (ξp), KSIn (ξn), and KSIv (ξv). At a minimum, the data sets will need to include: VS, RTS, THDF, wTS, ETAR, ETAO, ETAD, PD and RPM.

Propeller-cavitation tests will need to be carried out at laden and ballast drafts, and the owner should ensure that the testing facility uses a methodology that follows procedures recommended by the International Towing Tank Conference.

To allow for a comprehensive systems analysis, owners may also consider requesting/sourcing additional data, including:

- Wind-resistance coefficients based on wind-tunnel testing at various drafts
- Full hydrostatic tables, including block coefficient, length waterline, wetted surface area, projected transverse areas above and below waterline, waterplane area, pitch radius of gyration to length ratio (kyy), and entrance angles of the fore and aft waterlines

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The following information can be retrieved from vessels' manuals, plans and drawings:

- Specified Maximum Continuous Rating (SMCR) Power and RPM
- RPM at higher and lower barred speed range
- Height of anemometer from keel
- Propeller diameter
- Propeller-pitch ratio
- Height of shaft above the baseline

The builder also could make additional weather-resistance and power calculations for the owner that are specific to drafts, speeds (as per the model tests) and different weather conditions, i.e., significant wave heights and peak wave periods. These include:

- Response amplitude operators
- Added wave resistance
- Added wind resistance
- Power and RPM in weather

Measurements examining the roughness of the hull could be provided to the owner prior to launch; fuel-oil tanks could be calibrated to determine the "as-built" configuration and any changes that were made during construction.

To avoid any discrepancies from the tank-volume tables being realized during service, especially for the intermediate levels under trim/list, the tables of all fuel tanks should be calibrated to improve the accuracy of any calculations.

Owners will be trying to transition their fleets in an era where "smart" technologies are increasingly available to gather and assess information. The more information that is inputted, the better the chances of arriving at the desired result. A key will be to ensure the quality of whatever data is collected.

ABS SUSTAINABILITY NOTATIONS

In September 2015, the United Nations (UN) adopted its 2030 Agenda for sustainable development, which included 17 Sustainable Development Goals (SDGs). The Agenda emphasized the need to simultaneously consider three dimensions of sustainable development: social, economic and environmental. With the IMO actively working to align with the 2030 Agenda and its SDGs, shipping has faced some challenges to meet those goals in a safe, sustainable and cost-effective manner.

In recognition that marine operators require support to align with the SDGs, ABS last year introduced two new sustainability notations: SUSTAIN-1(2020) and SUSTAIN-2(2020). Having these notations assigned provides strong testament to financiers and charterers about the sustainability of the fleet and the company. The following chart presents the notations' key items and highlights their alignment with the UN's SDGs.

ITEM	TOPIC	SDG	SUSTAIN-1 (2020)	SUSTAIN-2 (2020)
1	Oil and Chemical Pollution	14 LIFE BELOW WATER; 3 GOOD HEALTH AND WELL-BEING	✓	✓
2	Waste Streams	12 RESPONSIBLE CONSUMPTION AND PRODUCTION; 14 LIFE BELOW WATER	✓	✓
3	Coastal and Marine Ecosystems	14 LIFE BELOW WATER	✓	✓
4	Air Emissions	3 GOOD HEALTH AND WELL-BEING; 7 AFFORDABLE AND CLEAN ENERGY; 9 INDUSTRY, INNOVATION AND INFRASTRUCTURE; 13 CLIMATE ACTION	✓	✓
5	Efficiency and Performnace Monitoring	7 AFFORDABLE AND CLEAN ENERGY; 9 INDUSTRY, INNOVATION AND INFRASTRUCTURE; 13 CLIMATE ACTION	✓	✓
6	Ship Recycling	3 GOOD HEALTH AND WELL-BEING; 12 RESPONSIBLE CONSUMPTION AND PRODUCTION	✓	✓
7	Low-Carbon Fuels	7 AFFORDABLE AND CLEAN ENERGY; 13 CLIMATE ACTION		✓
8	Human-CenteredDesign	3 GOOD HEALTH AND WELL-BEING; 8 DECENT WORK AND ECONOMIC GROWTH		✓

The "Oil and Chemical Pollution" item covers the machinery spaces and cargo areas of oil tankers and the carriage of noxious liquid substances; the "Waste Streams" item covers sewage and garbage; and the "Coastal and Marine Ecosystems" item mainly covers ballast water, antifouling and biofouling management systems, as well as the effects of noises radiated under water. The "Air Emissions" item includes requirements for dealing with ozone-depleting substances.

More specifically, the SUSTAIN-1(2020) notation can be assigned (subject to meeting the rest of the requirements) to any tanker that will comply with the upcoming EEXI regulations and any new tanker that exceeds the requirements of their EEDI reduction (shown in Table 2) against the EEDI Phase 0 baseline.

Additionally, if any of the "innovative technologies" mentioned in the Key Design Features for the Next Generation of Tankers chapter have been adopted, they must be indicated in the EEDI Technical File, or their implementation verified as "Energy Efficiency Technologies" in accordance with MARPOL Annex VI, Chapter 4. Vessels are also required to have a class-approved SEEMP, an operational measure that establishes a mechanism to improve the energy efficiency of a ship in a cost-effective manner.

TANKER	≥ 20,000 dwt	22	15
	< 200,000 but ≥ 20,000 dwt	2	20
	< 20,000 but ≥ 4,000 dwt	0-22*	0-20*

Table 2: Required EEDI reductions for tankers.

Vessels complying with the requirements of the SUSTAIN-1(2020) notation also may be assigned the notation SUSTAIN-2(2020). The latter notation can be assigned to tankers equipped with single- or dual-fuel engines or alternative power-generation systems (e.g., fuel cells, hybrid-electric power systems, etc.) for main propulsion that are designed to use low- or zero-carbon fuels, including: LNG/CNG; LPG; ethane; methanol; biofuels; ammonia; and hydrogen. For more information on these notations please refer to the ABS *Guide for Sustainability Notations*.

ABS ALTERNATIVE FUELS READY PROGRAM

In February 2021, ABS introduced the "Alternative Fuel Ready" program and supporting notations to address the ever-increasing environmental measures described in the first chapter on regulation. It is widely believed that the adoption of LNG, methanol, ethane, LPG, hydrogen, ammonia and other gases or low-flashpoint fuels is the most viable strategy to achieve long-term compliance with the majority of the current and upcoming regulations.

Designing a ship that can be later converted to alternative fuel use can limit the initial investment for a new tanker, while holding the option to select gases or other low-flashpoint fuels in the future. This arrangement is generically known as "Alternative Fuel Ready." The scope of such preparation or modifications significantly differs from ship to ship, so it needs to be agreed between the shipowner and the shipbuilder on a case-by-case basis.

ABS will consider the application of the "Alternative Fuel Ready" program and notations to tankers falling under the scope of the International Code of the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk (IGC Code) on a case-by-case basis, provided such proposals are arranged in accordance with the requirements of the Code and 5C-8 of the Marine Vessel Rules, and with agreement of the flag Administration. There are three fundamental levels that define the readiness of a vessel seeking certification under the program, including:

- **Level 1 — Concept Design Review:** a high-level evaluation of compliance with the Marine Vessel Rules and the basic suitability of a vessel design for gas or other low-flashpoint fuel concepts.
- **Level 2 — Detail Design Review:** Additional to Level 1, it is categorized into separate groups to identify the parts of the design that are reviewed for compliance with the Marine Vessel Rules.
- **Level 3 — Installation:** The final level of the program that extends from class approval of the drawings to the installation of parts of the system and specified equipment on board the vessel, including survey in accordance with the requirements of the Marine Vessel Rules.

For more information on these notations please refer to the ABS *Guide for Gas and Other Low-flashpoint Fuel Ready Vessels*.

LIST OF SERVICES OFFERED BY ABS

ABS can assist tanker owners, operators, shipbuilders and original equipment manufacturers with the following services:

- Risk assessment
- Regulatory and statutory compliance
- New technology qualifications
- Life-cycle and cost analysis for ammonia-fueled vessels
- Vessel/fleet benchmarking and identification of options for improvement
- NO_x emission-reduction options
- EEDI verification and identification of options for improvement
- Optimum voyage planning
- Alternative fuel-adoption strategy
- Technoeconomic studies
- Cyber safety notations and assessments
- Contingency arrangement planning and investigation

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